



life.augmented

ST Smart Industry Robotics solutions V9

Robin Feng

Strategic Marketing Manager

SMART INDUSTRIALS Segment



Agenda

1 Robotics system

2 Visualization

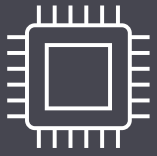
3 Processing& Control

4 Connectivity

5 Actuator

6 Sensing

7 Power



Robotics Systems



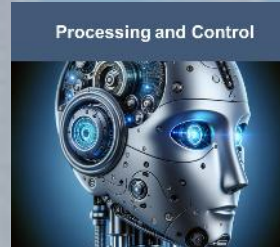
“Mid-Long Term Attack Plan”

Leverage on Automation Systems knowhow and Address Robotics system painpoints

Co-Define Product Definition@system: Robotics System



Visualization



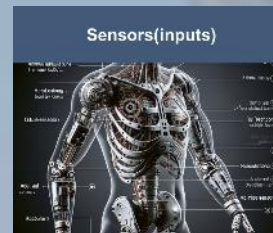
Processing & Control



Connectivity



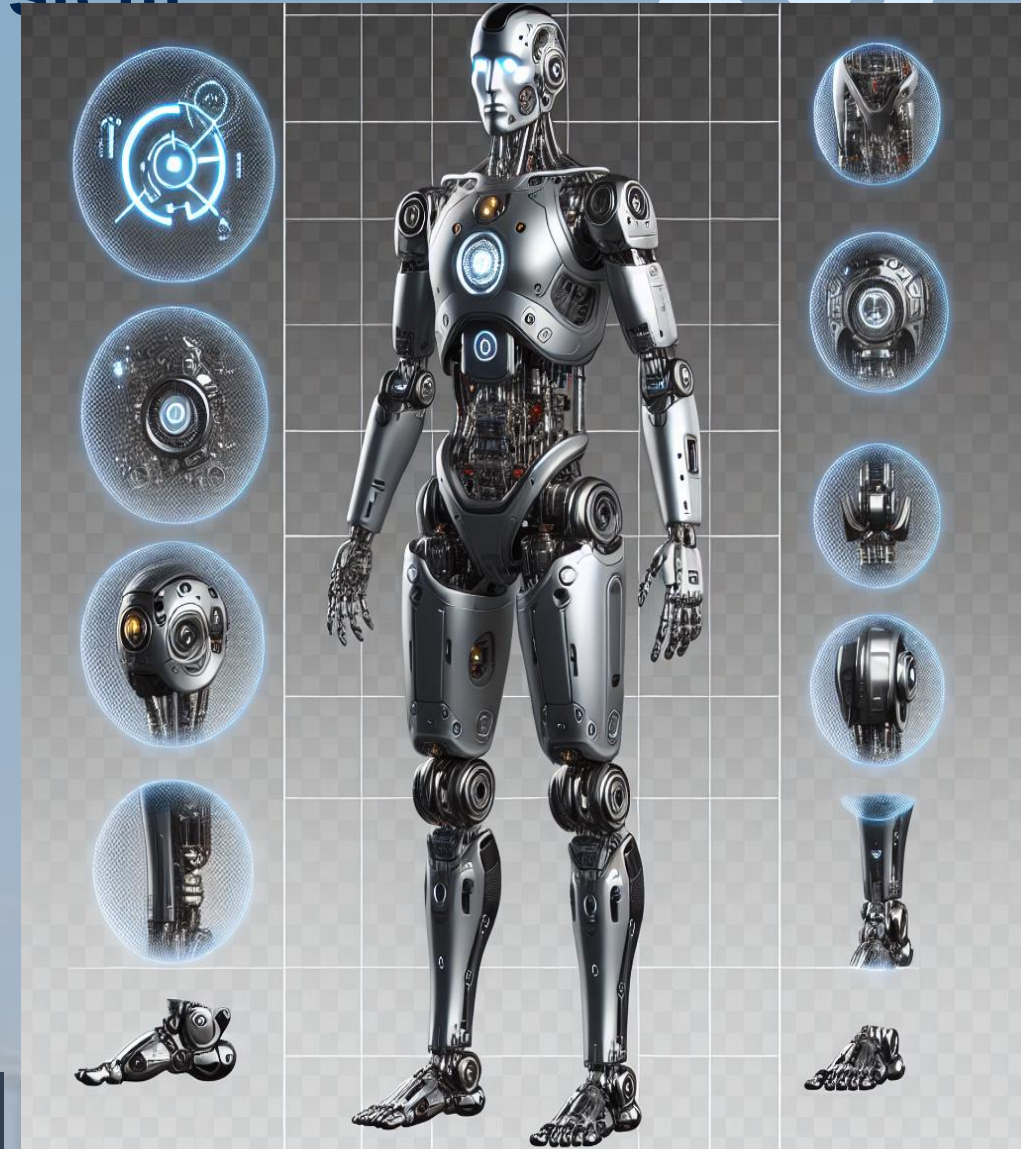
Actuator



Sensing



Others



Translating AI-Language to Automation Language

21 finger points via LVTTTL-RS485-PLCcodesys-CANFD to 15 Servo motors

WE support Market Trend: like Artificial Intelligence

21 finger key points' data (X, Y, Z)



PLC Data Processing

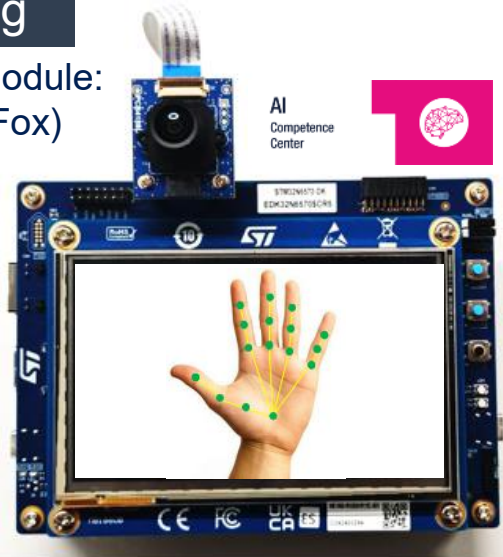


15 servo motors' position data

Sensing

Camera Module:
VD66GY(Fox)

AI
Competence
Center



STM32N6 EVM Board

Others: AI

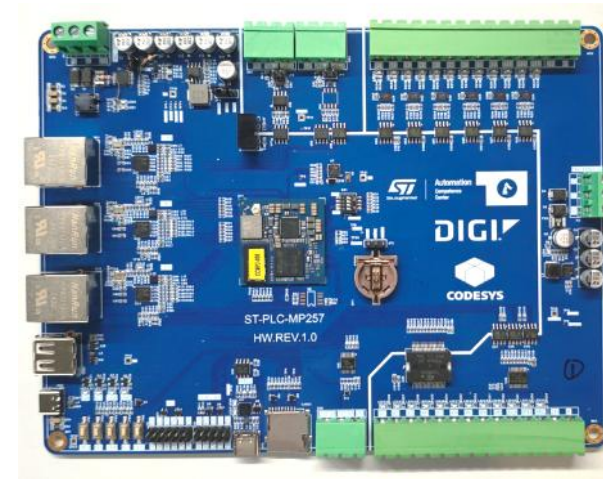
ST Demo:
stm32n6570-dk

Connectivity

UART



RS-485



STM32MP257 PLC Board

Processing & Control

ST DEMO:
stm32mp257f-ev1

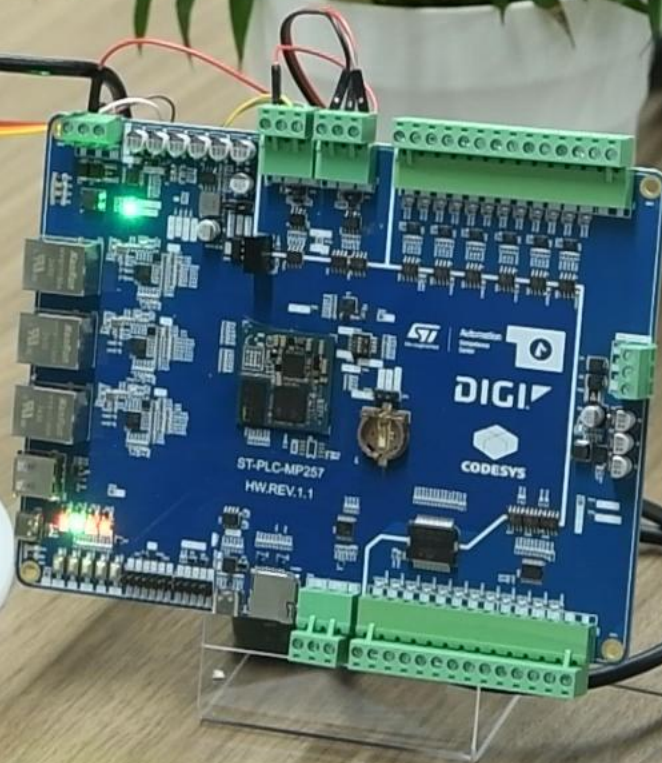
CANFD



Robot Hand

Actuator

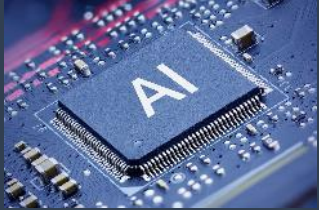
Ruiyan Co-development
RY-H2R-0611-C0-ST-00
RY-H2L-0611-C0-ST-00



Robotics System Platform: (System Building Blocks)

AI: STM32N6

21 finger key points' data (X, Y, Z)

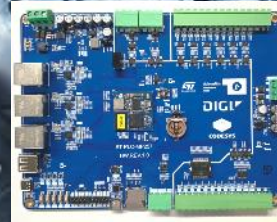


Visualization

Sensing

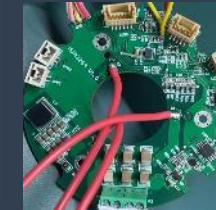


Connectivity



Processing &
Control

PLC Data Processing



Actuator



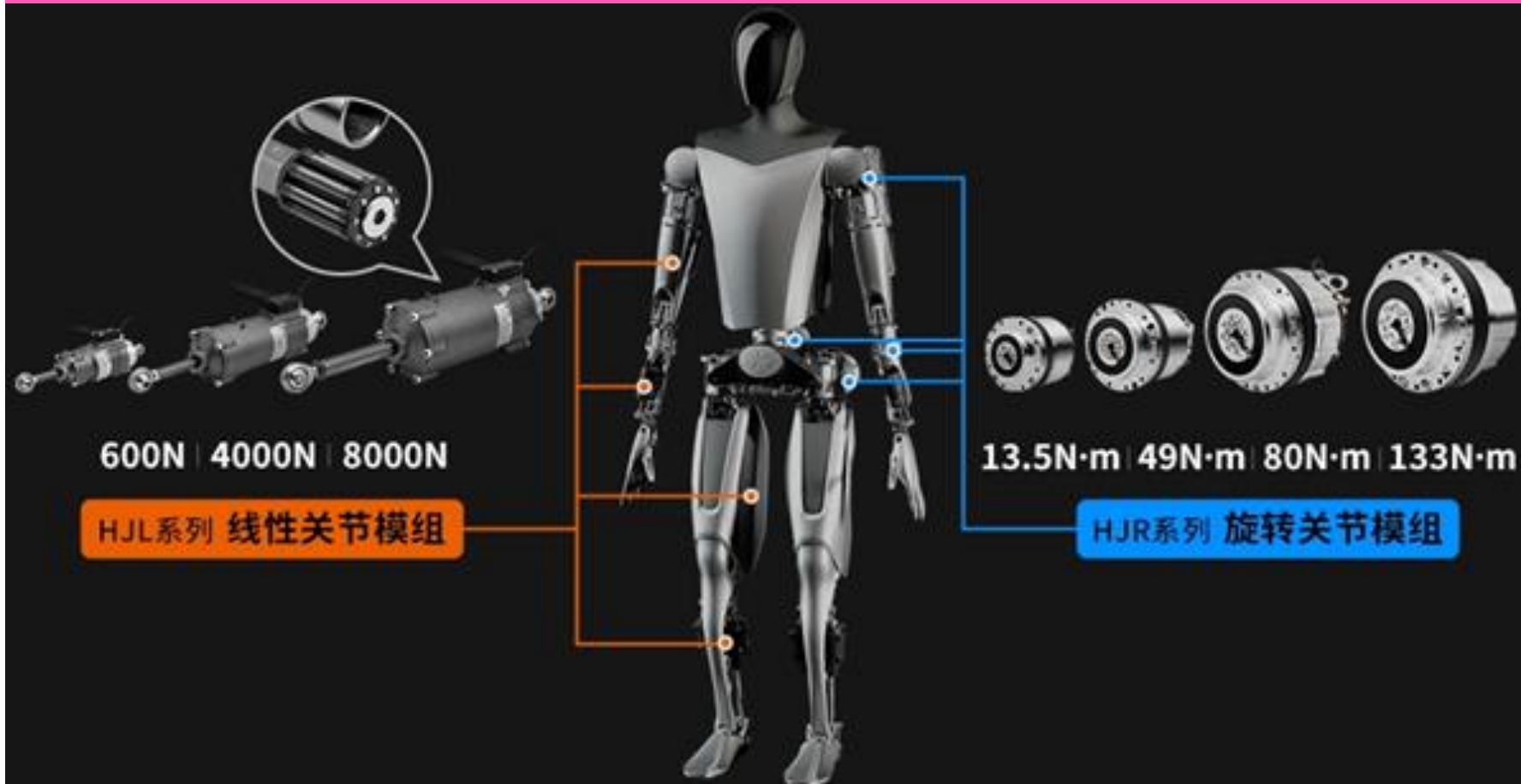
15 servo motors' position data

Humanoid Robot System Breakdown:

- **Total 44 +Motors with controller in each robot.**
- Head: Camera, Laser Lidar, AI processor, Sensor;
- Body+ two Hands: 14 Motors + Controller, Sensor;
- Two Legs: 14 Motors+ Controller, Sensors;
- Two Smart Palm: 5-15 Servo Motors+ Controller, Tactile Sensors, Touch sensor;
- Communication module: BLE, Wifi, RF, 2.5GHz, 5GHz..
- Battery Power 48~52V.

Legs and Hands: Servo Drive

There are two kind Joint movement module:
Prismatic Joint module and Revolute Joint module



Humanoid Robot Breakdown

Head



- 3*Camera 150\$
- Lidar/Radar 100\$
- AI processor 1400\$
- Other Sensors 2000\$

2*Smart Hand



- 15*Motor 1200\$
- 15*Gears 500\$
- 15*Ballscrew 350\$
- 15*Touchsensor 2000\$
- 15*Torque sensor 350\$
- 2*IR sensor 150\$

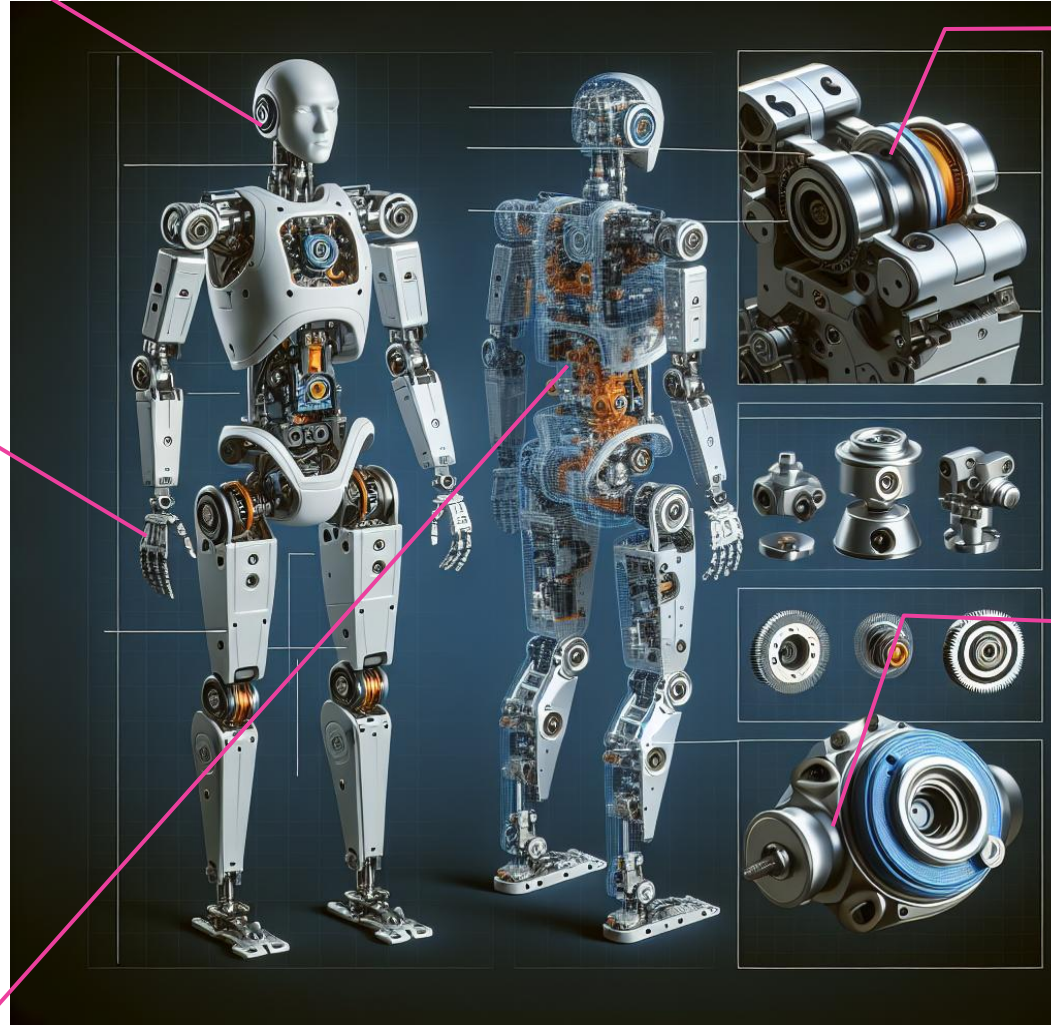
Body



1*Battery Power 700\$



1*Connectivity 1000\$



Arm: 14*Revolute Joint



- 14*Harmonic Gear: 1400\$
- 14*Motor+Driver: 3000\$
- 14*Torque Sensor: 400\$
- 14*Bearing: 100\$

Leg: 14*Prismatic Joint



- 14*Motor+Driver: 3000\$
- 14*Torque Sensor: 400\$
- 14*Harmonic Gear: 2000\$
- 14*Bearing: 100\$

Visualization (HMI/Display, Vision...)

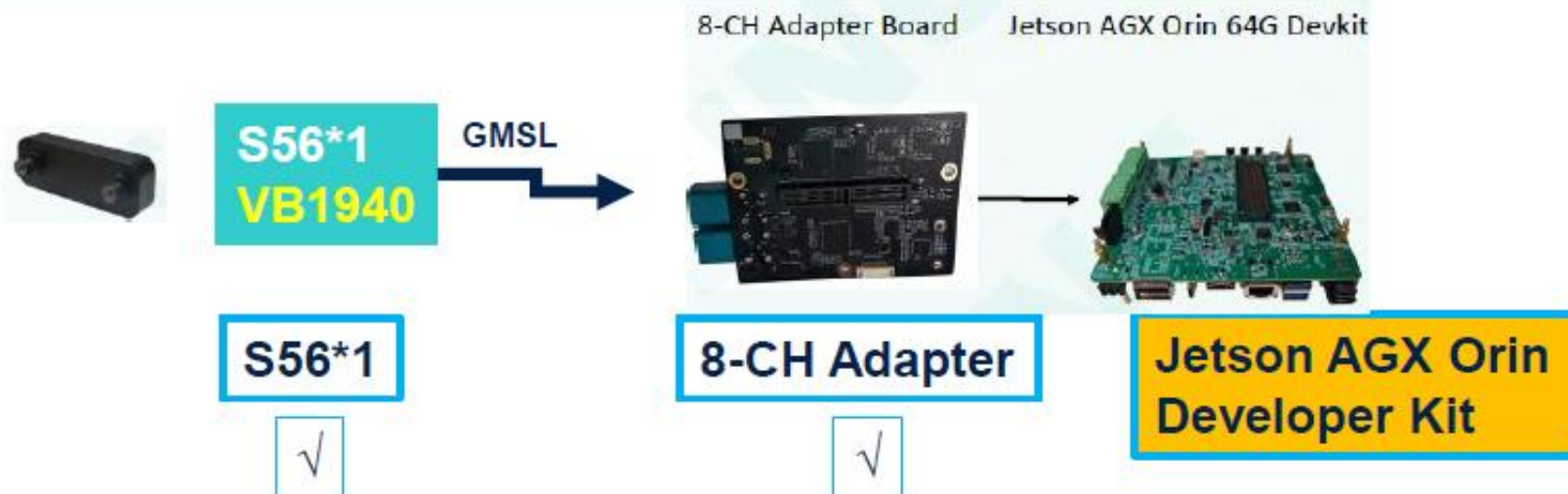


HSB and Camera solution from Sensing

Scheme based on Holoscan Sensor Bridge

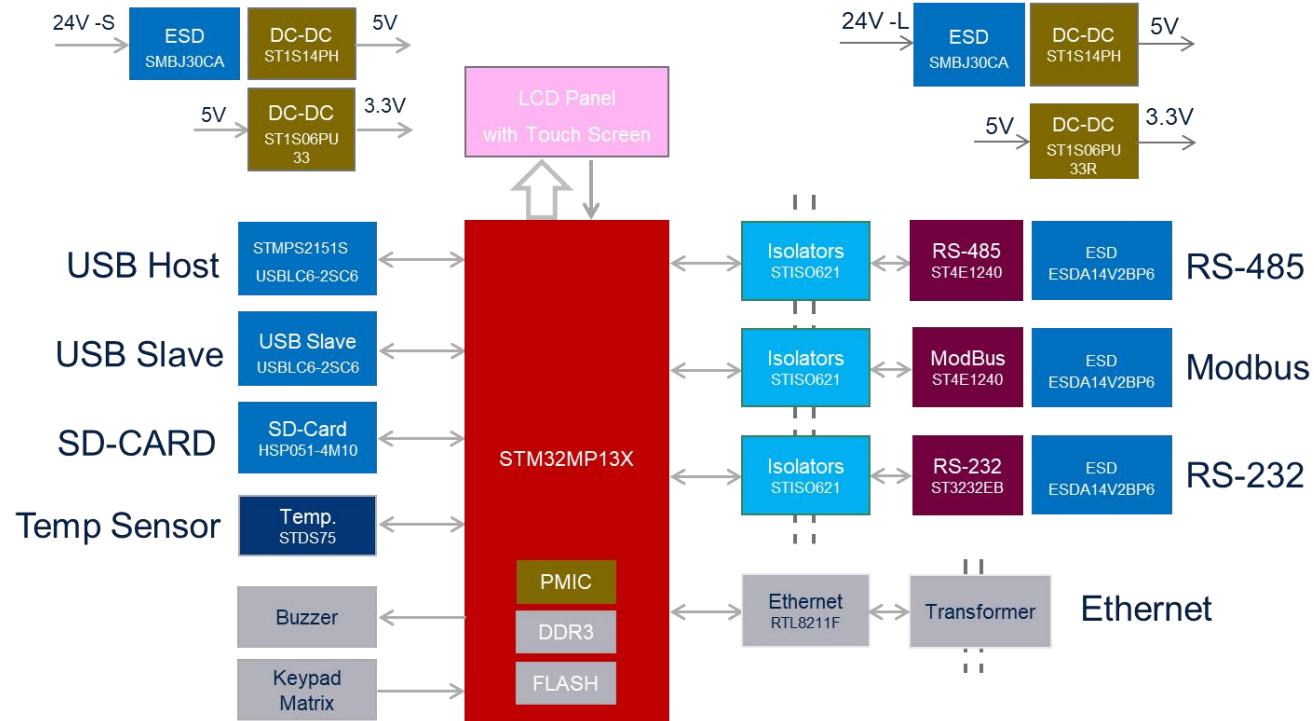


A universal humanoid vision solution based on the Jetson Orin carrier board



Human Machine Interface: Reference Design

HMI reference design(expandable and configurable) platforms:



Human Machine Interface



8-10 inch Touch Screen HMI
Custom-development for a System Integrator in India

Processing and Control





ST Automation PLC system Solution Matrix

Hardware
Platform



STM32MP2



STM32MP1



STM32H7



Software
Platform



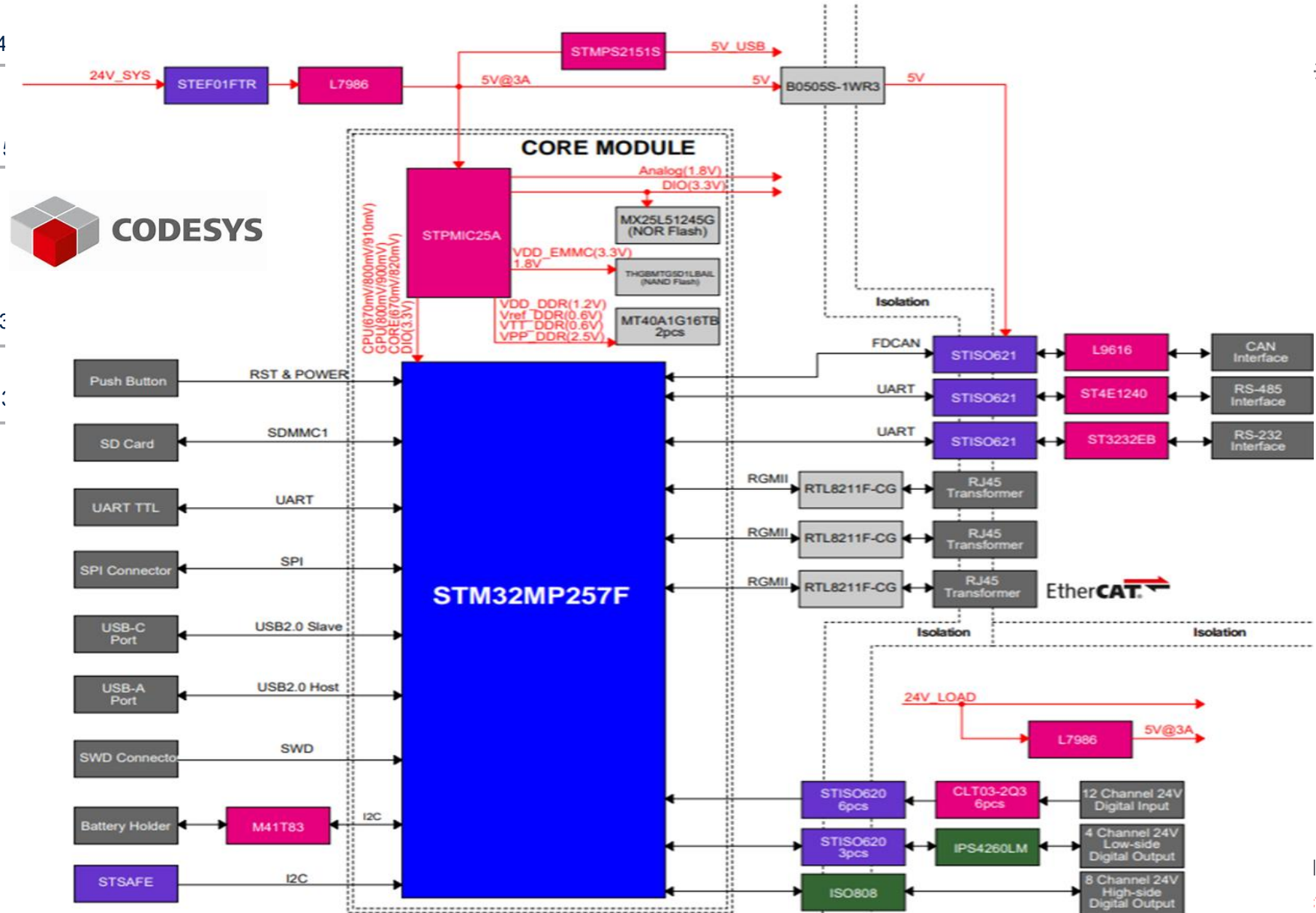


ST PLC+Motion control reference design

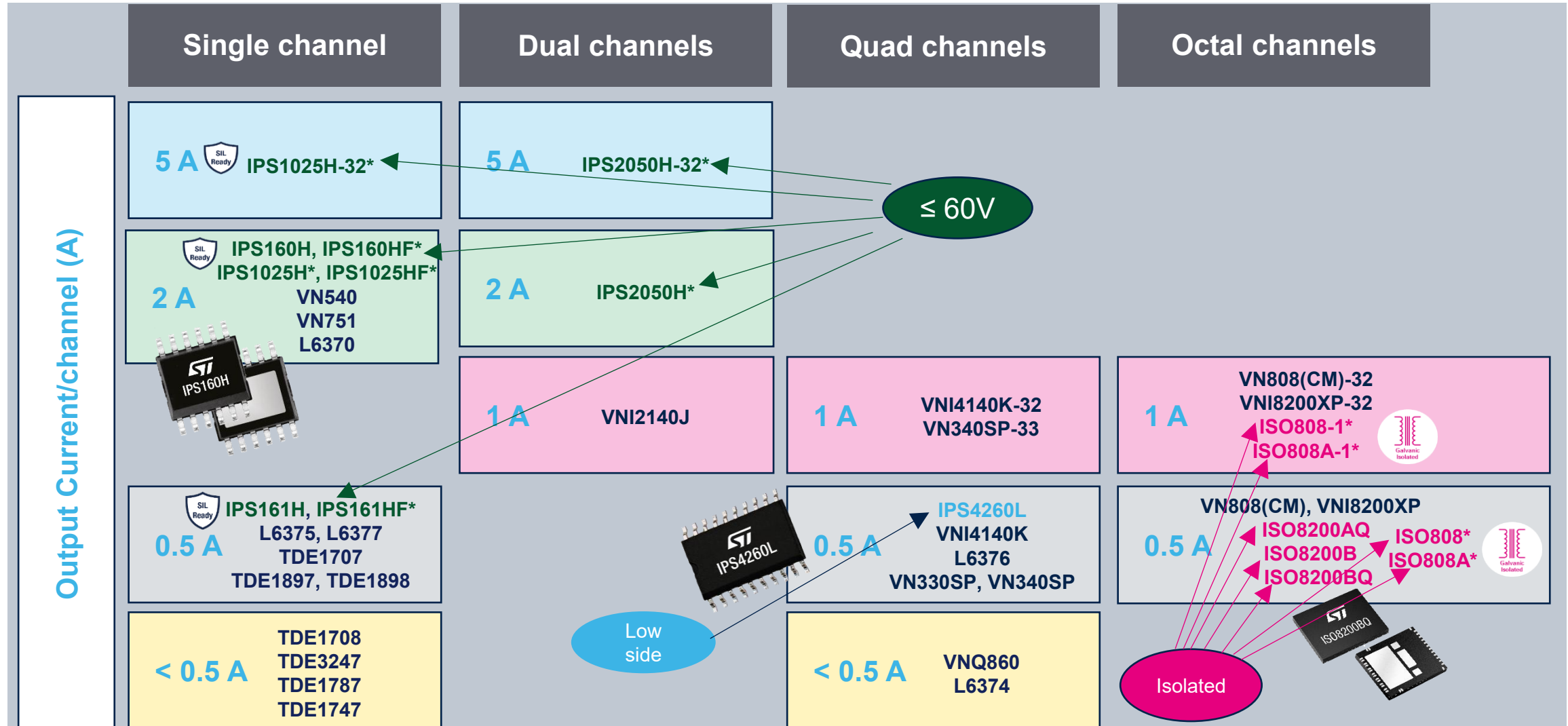


- High performance STM32MP257 MPU
- 8-ch isolated high-side IPS for DO
- 4-ch isolated low-side IPS for DO
- 8-ch isolated digital input
- 3 Ethernet port support EtherCAT controller interface
- RS-485/ModBus CAN, RS-232
- USB, microSD™ card interface

24

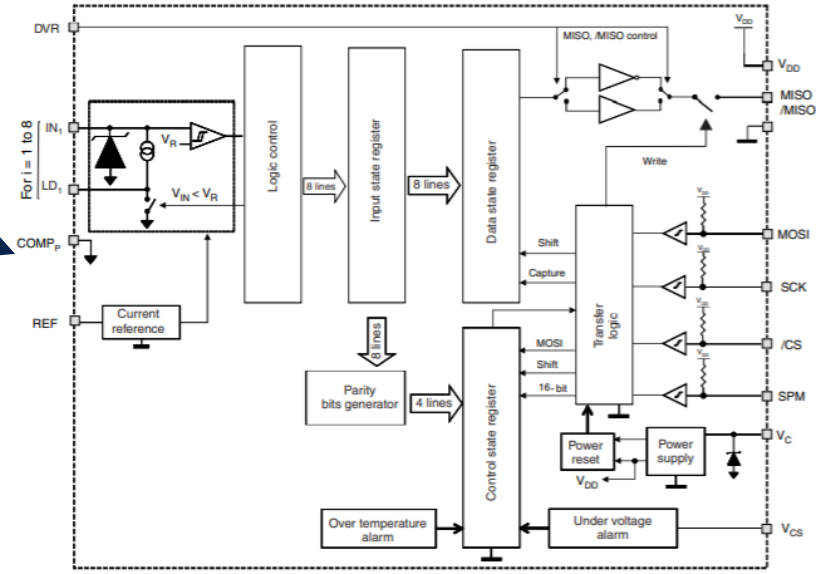
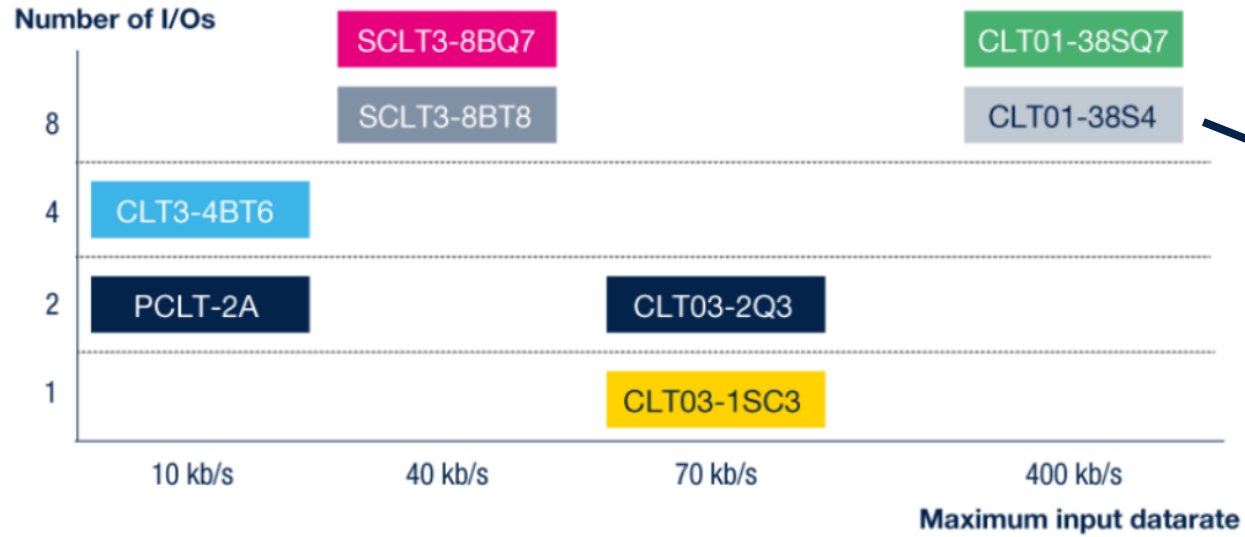


Intelligent power switch family



(*) in development

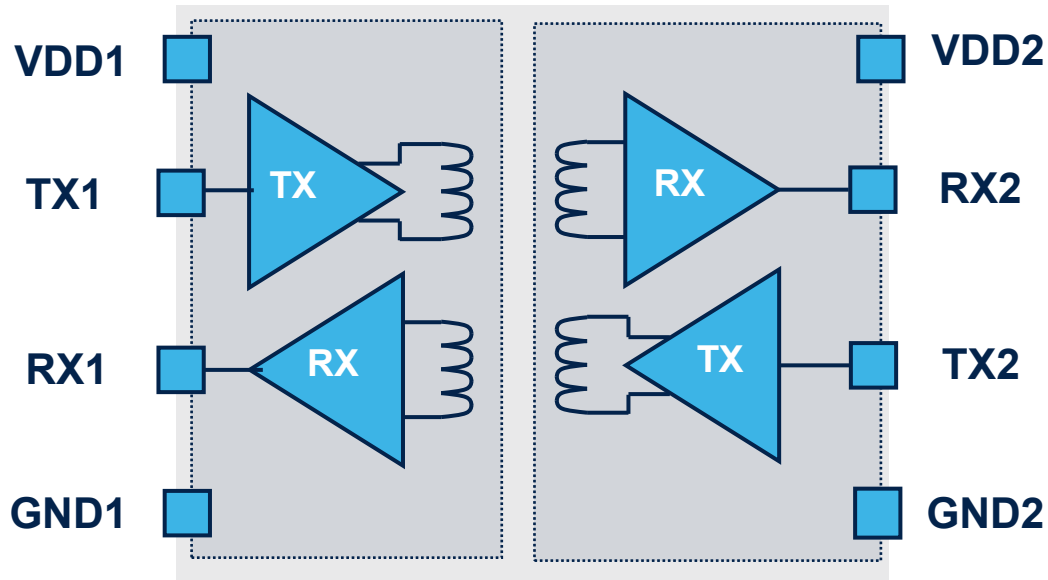
Current limiters for Digital input



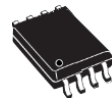
- Up to 8 protected integrated channels available
- Up to 2 MHz data interface speeds, 400kbps input rate, SPI available
- 600 W overvoltage protection integrated
- Highly-immune solutions such as those described in IEC 61000-4
- Power dissipation halved for more integrated modules

Galvanic Isolation Connectivity: STISO621/0 - 6KV Isolated UART Interface

Low Power, high speed UART I/F



- SO8W In Production
- SO8N qualification Dec '20



SO8W



SO8N

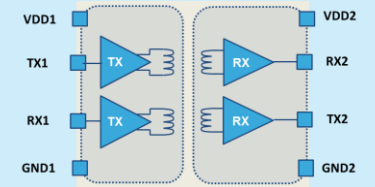
6kV galvanic isolation , 100Mbps rate and low pulse distortion (<3ns) for safe and fast data transfer between isolated domains

10mA power consumption at max data speed for ultra-low power and high performing operation

CMTI = 65kV/us typ, Propagation Delay 25 ns typ

High flexibility with 3.3V and 5V supply ratings and with SO8N and SO8-W package options

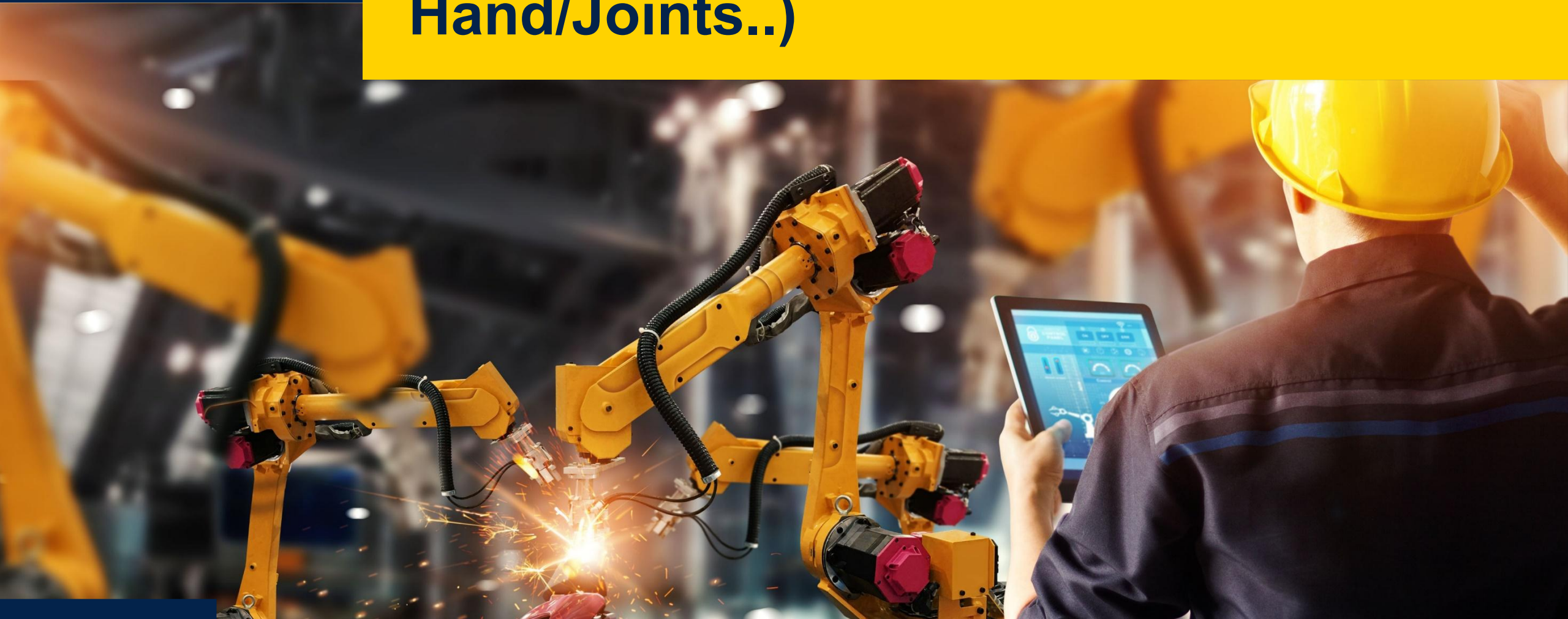
Dual channel, one direction
(**STISO620**) under development



<https://www.st.com/en/interfaces-and-transceivers/stiso621.html>

Prod. evalution board EVALSTISO62XV1 available

Actuator(Motor Control, Sound, Robot Hand/Joints..)



Humanoid Robots: Major Joints

"Robotics SSP target: A full modular ST platform to show leadership in Robotics"

Head Neck Joint: 2~4 BLDC Motors + ESC

Precise movement,
Smooth motion
BLDC,PMSM,
Low Gear ratio 10:1~50:1
Motor Parameter:
48V, 5~50W,
0.1~10Nm, 180~360°/s



Elbow Joint: 2 BLDC Motors +ESC

High Torque, Speed-accuracy tradeoff
BLDC Servo Motors,
Middel Gear ratio:20:1~100:1
Motor Parameter:48V 10~500W, 5~200Nm, 180~720°/s



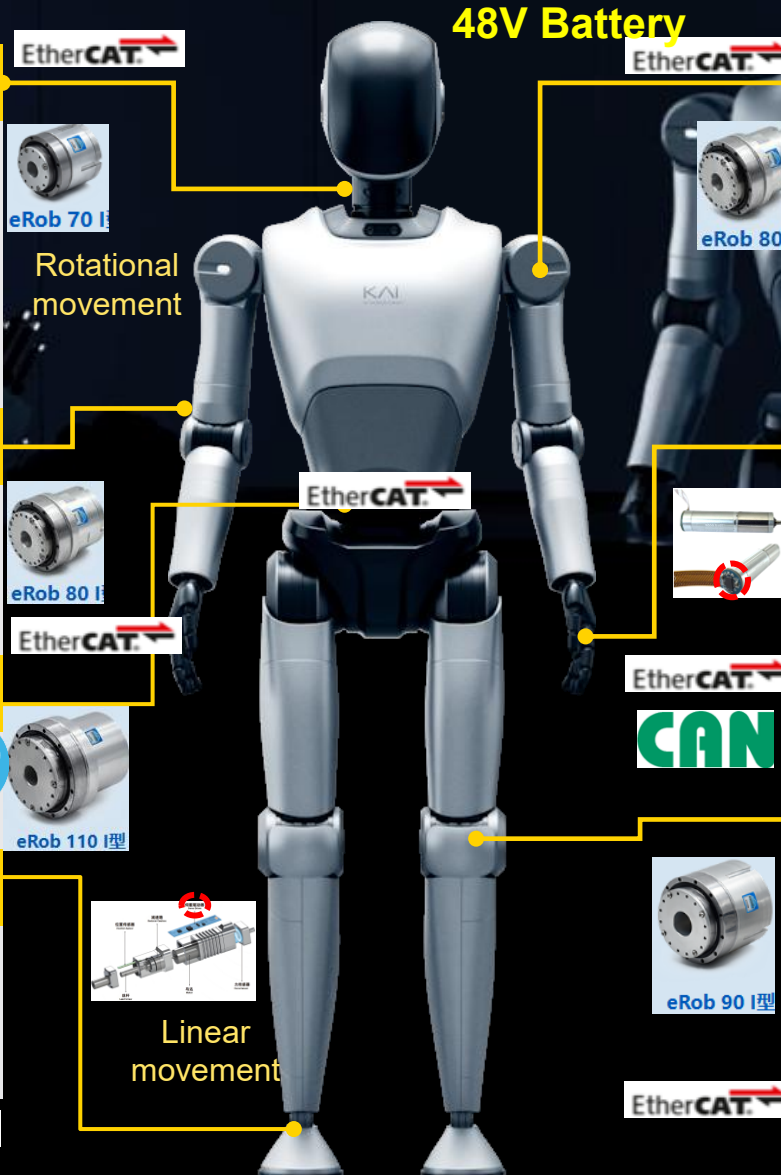
Hip and Waist Rotation Joint: 4 BLDC Motors

High Torque, shock absorption
BLDC,High Gear ratio: 80:1~150:1
Motor Parameter: 48V20~3KW, 10~500Nm, 180~540°/s



Ankle Joint: 4 BLDC Motors +ESC

Dynamic response for balance
BLDC Servo Motors,
Middel Gear ratio:40:1~100:1
Motor Parameter:48V 10~500+W, 5~200Nm 180~360°/s



Shoulder Joint: 2 BLDC Motors +ESC

High Torque,
Speed-accuracy tradeoff
BLDC Servo Motors,
Middel Gear ratio: 20:1~100:1
Motor Parameter: 48V 10~500W,
5~200Nm, 180~720°/s



Hands unit: 6~22 coreless DC Motors + ESC or CAN

Compact size,
Low weight,
Accurate,
Dexterous
Codeless DC Motor,
mimic gear ratio /direct drive
Motor Parameter: 12/24V1~10W,0.1~2Nm,
360~1080°/s



Knee Joint: 2 BLDC Motors

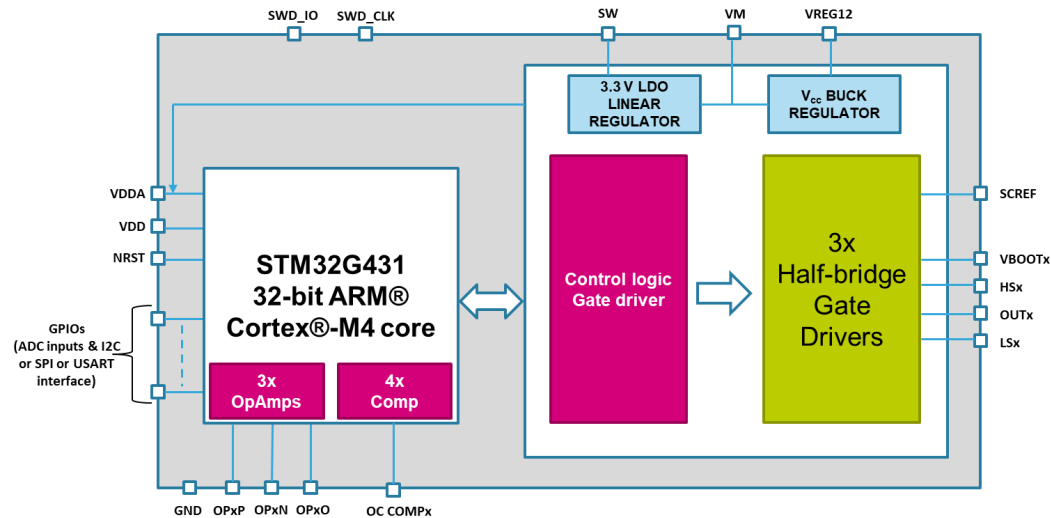
High Torque,
Shock absorption
BLDC,PMSM,
High Gear ratio: 80:1~150:1
Motor Parameter:48V 20~3KW, 10~500Nm,
180~540°/s





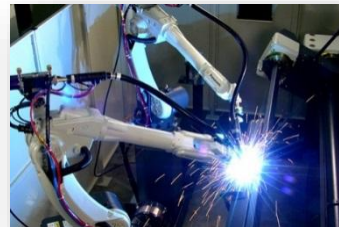
STSPIN32G4

Advanced 75V 3-phase BLDC Driver with embedded Cortex M4 STM32 MCU



KEY APPLICATIONS:

- Industrial automation & Robotics
- Battery powered Home Appliances
- Power tools
- Drones and aeromodelling
- Fans and pumps



Key Features

- **STM32 Cortex M4 + 3-phase Driver**
- **Fully compatible with STM32 ecosystem**
- **6step & FOC sensorless / sensed algorithms**
- **V_{cc} (200mA) Buck & 3.3V (150mA) Linear reg**
- VS = 5.5V – 75 V, I = 1A driving capability
- 170 MHz, 128k Flash & 32k SRAM
- 2x 16 bit ADCs & 4x 12 bit DAC channels
- I2C / UART / SPI / CAN; **up to 40GPIOs**
- **Low-power in stand-by**
- **MOSFET V_{DS} Monitoring protection**
- Up to 4x Comparators & 3x OpAmps
- Fully protected (UVLO, Short-circuit, OCP)
- Fully configurable thru I2C
- 9x9 QFN package



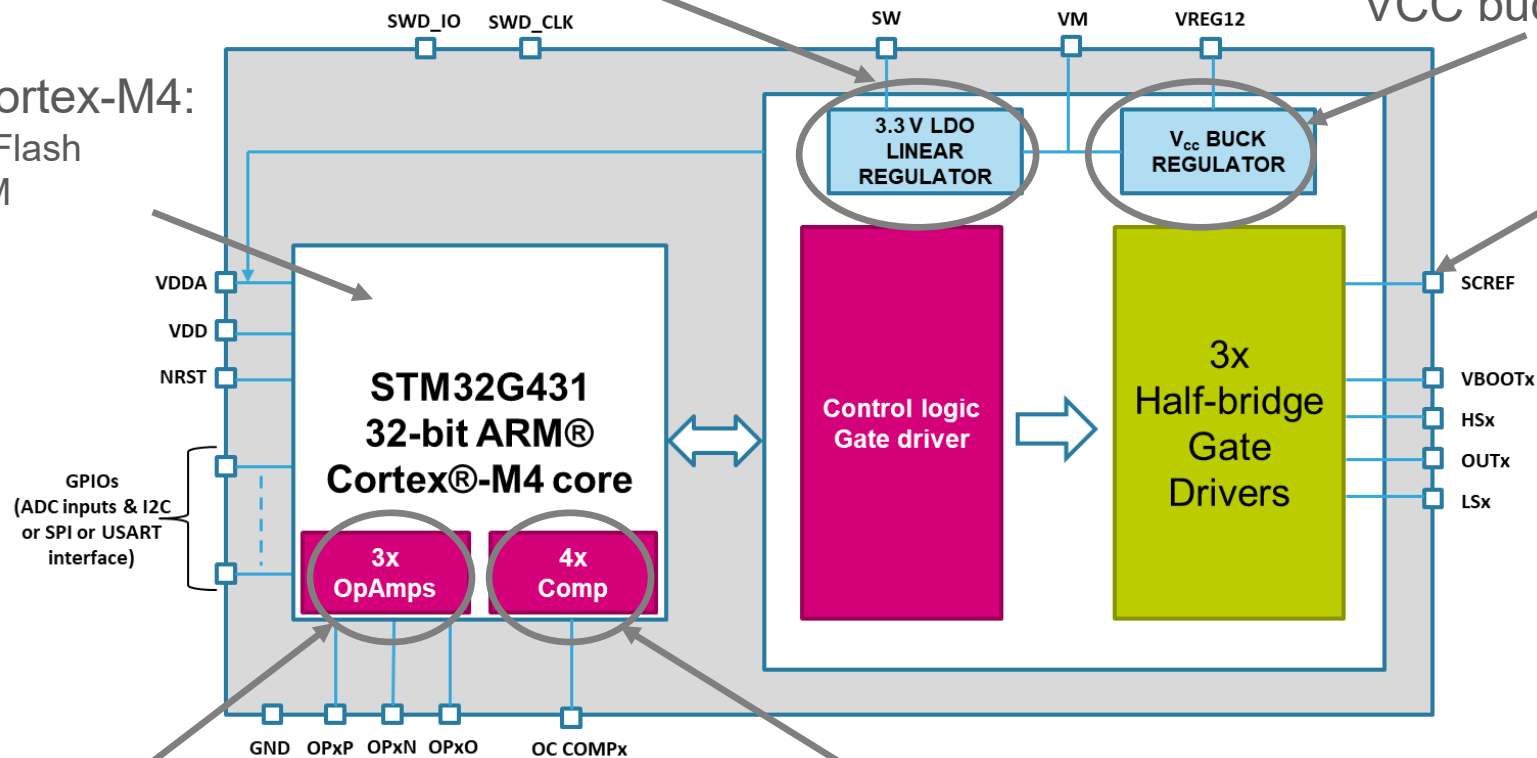
Advanced 75V 3-phase BLDC Driver with embedded Cortex M4 STM32 MCU

3.3 V LDO linear regulator up to 150 mA / 0.5 W

VCC buck converter up to 200 mA

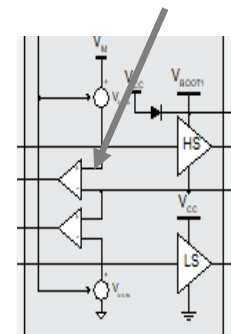
Mainstream Arm Cortex-M4:

- 128 Kbytes of Flash
- 32 Kbytes RAM
- 170 MHz CPU
- 4x USART
- timers
- ADC
- DAC
- comm. I/F
- 1.7-3.6V



Desat (SCREF) pin:
Ref V (0.2V-2.5V) for
embedded short-circuit
protection which monitors
external MOS V_{DS} :

$$V_{DStH} = V_{SCREF}$$



3x rail-to-rail operational amplifiers

15 MHz bandwidth

Rail-to-rail input/output

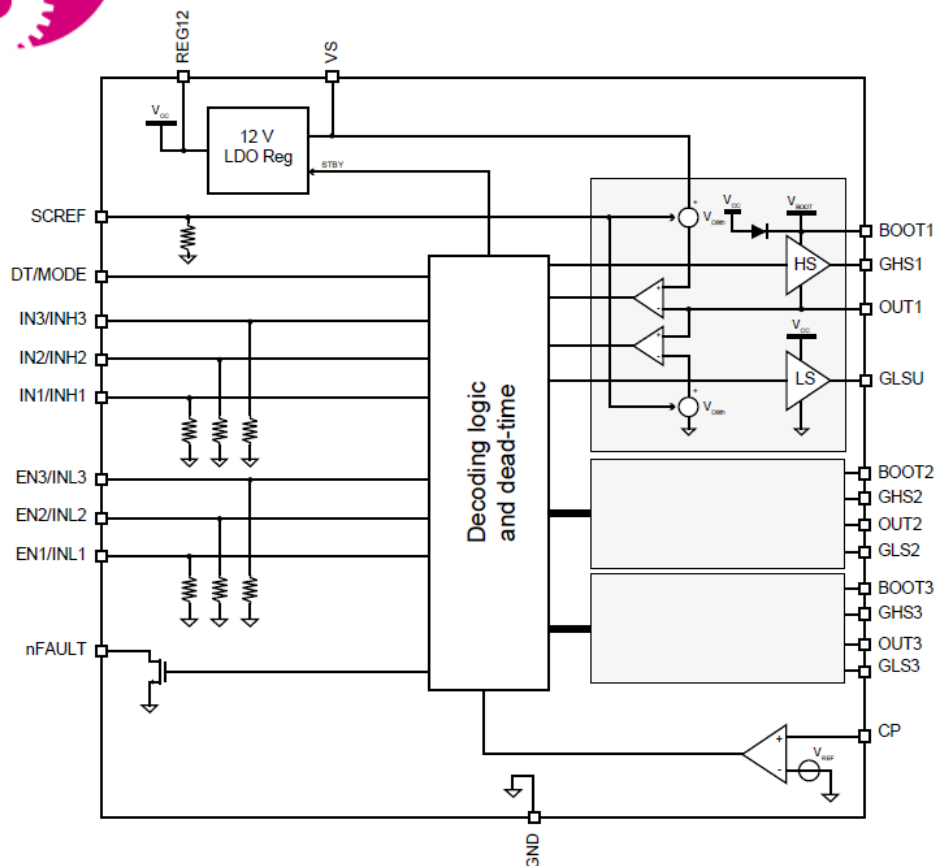
PGA (2, 4, 8, 16, 32, 64, -1, -3, -7, -15, -31 or -63)

4x fast comparators ($t_D < 35$ nS)



STDRIVE101

3phase BLDC Gate Driver

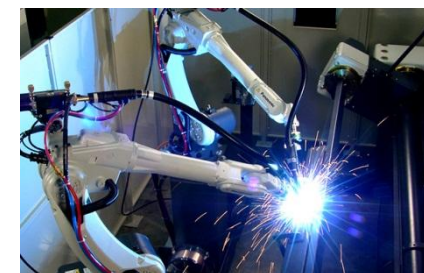


Key Features

- Operating Voltage: 5V to 75V
- $I = 600\text{mA}$ sink/source driving capability
- 12V LDO embedded
- Programmable Dead Time
- Desaturation (MOS Short Circuit) protection; programmable
- Fully protected (UVLO, OCP, thermal shutdown)
- Very short propagation delay 70ns; 20ns phases matching
- Compact design with 4x4mm QFN
- Integrated bootstrap diodes
- Standby mode for low consumption

KEY APPLICATIONS

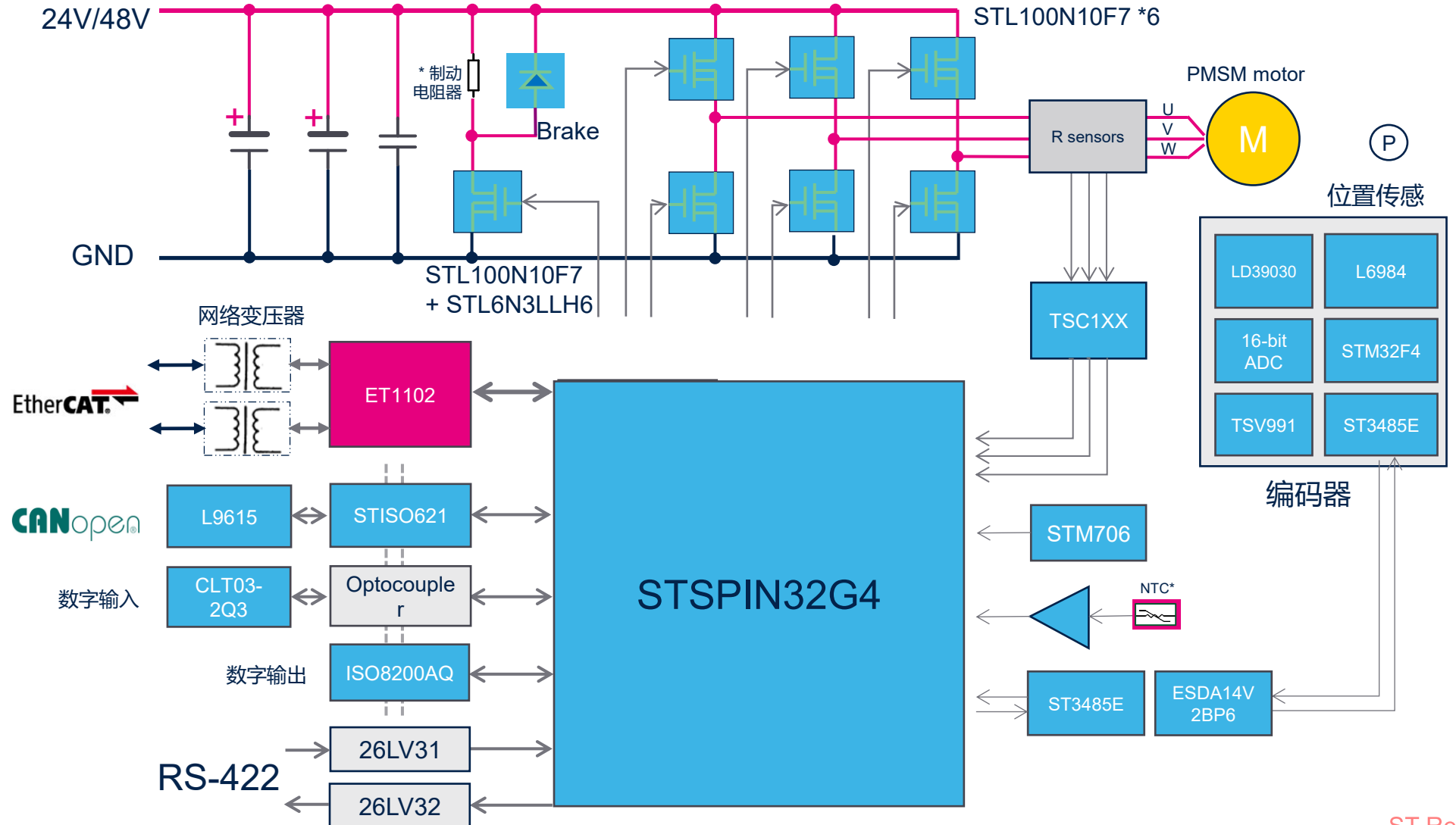
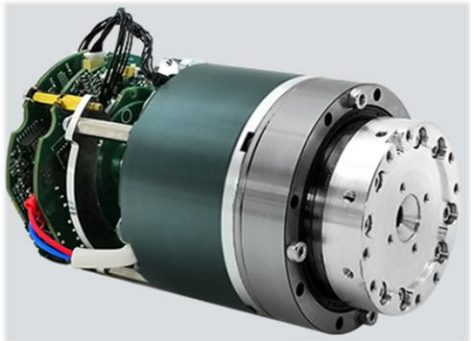
- Home appliances
- Power tools
- Factory automation
- Fans and pumps
- Robotics





Servo driver in robotic arm

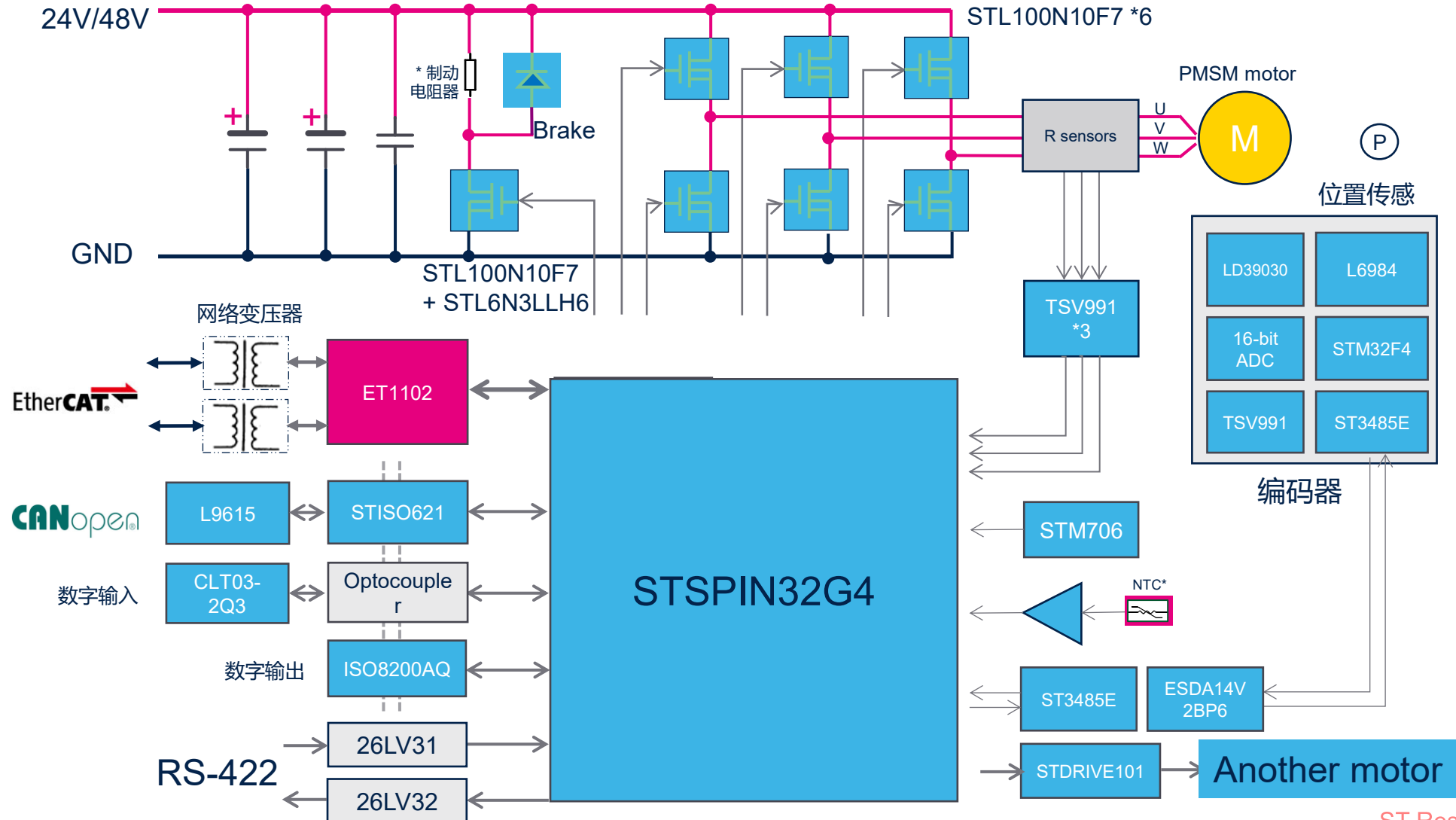
Industrial actuator demo system: robot arm



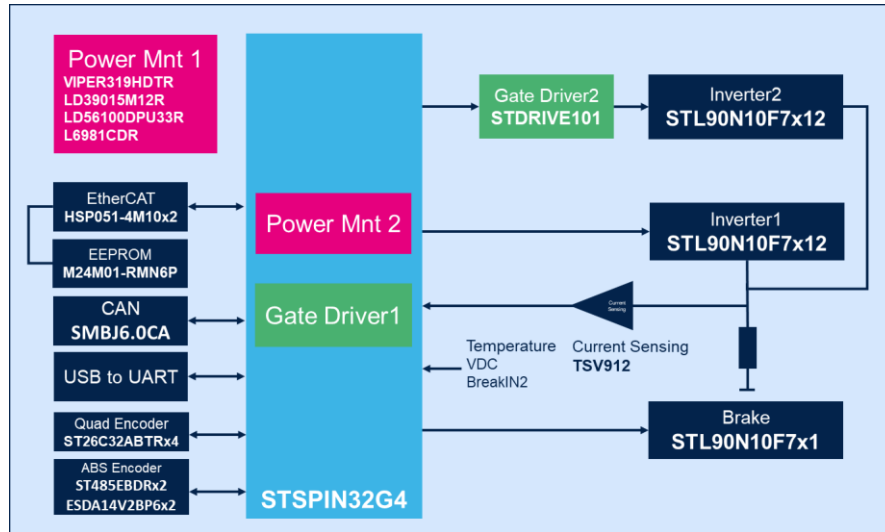


Servo driver in robotic arm

Industrial actuator demo system: robot arm



STSPIN32G4 Dual Drive Solution V2



Specs:

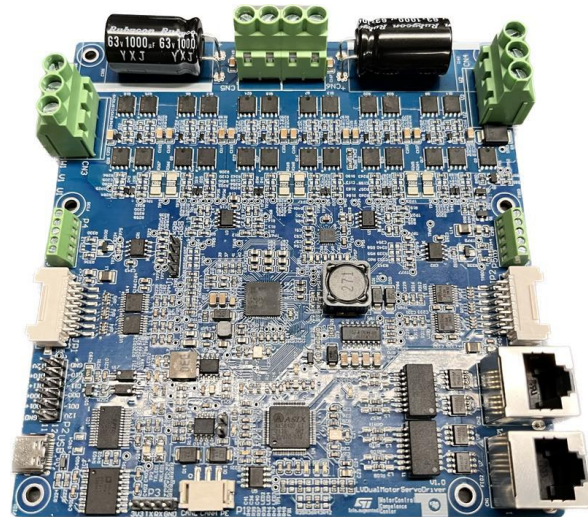
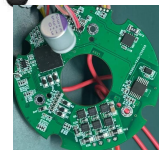
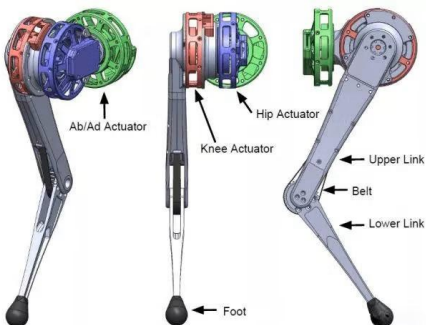
- 130x130mm dimension
- 1000W output power each
- 20 ~ 60VDC supply, 20Arms rating, 50A peak

Features:

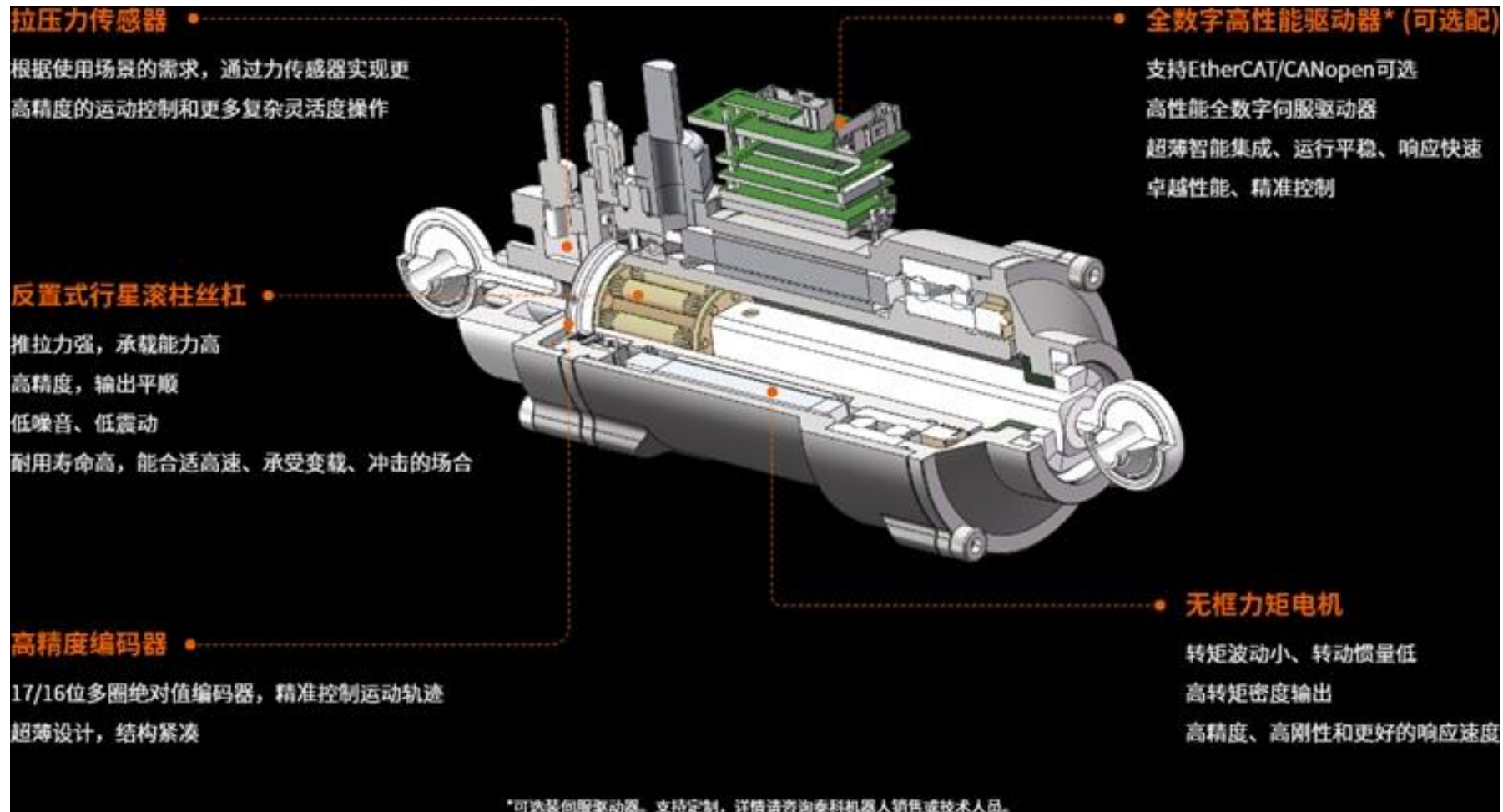
- Shared current sensing network
- 2 Switches in parallel per switch arm
- EtherCAT or CAN Onboard
- RS232 Onboard
- Multiple selectable configuration:
 - Dual differential ABZ+Hall encoder
 - Dual ABS encoder
 - Isolated digital output x2
 - Isolated digital input x2

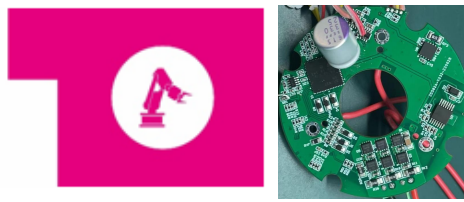
Applications:

- Robotic, AGV or other position control applications

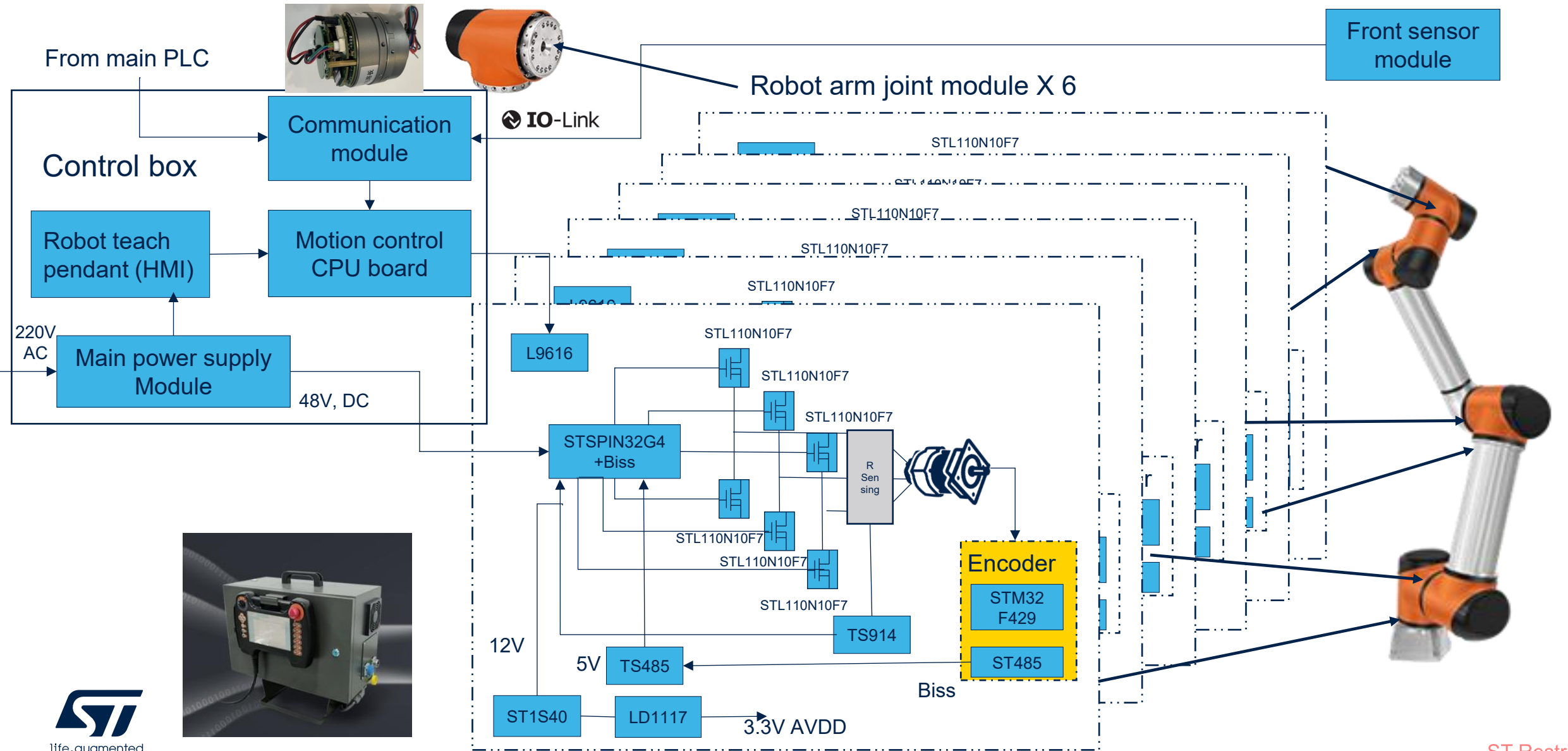


Prismatic Joint module





Joint servo module solution for robot arm

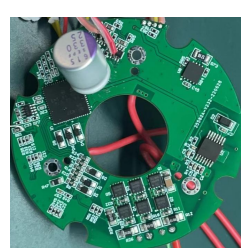


5Y PVO 14M\$

Solution Highlights

STSPIN32G4+Encoder for Robotics Servo Joint Module

MC.MD_0012.22



Key Features

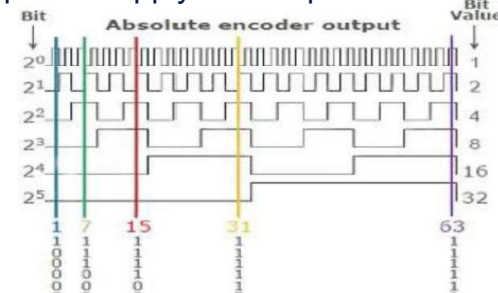
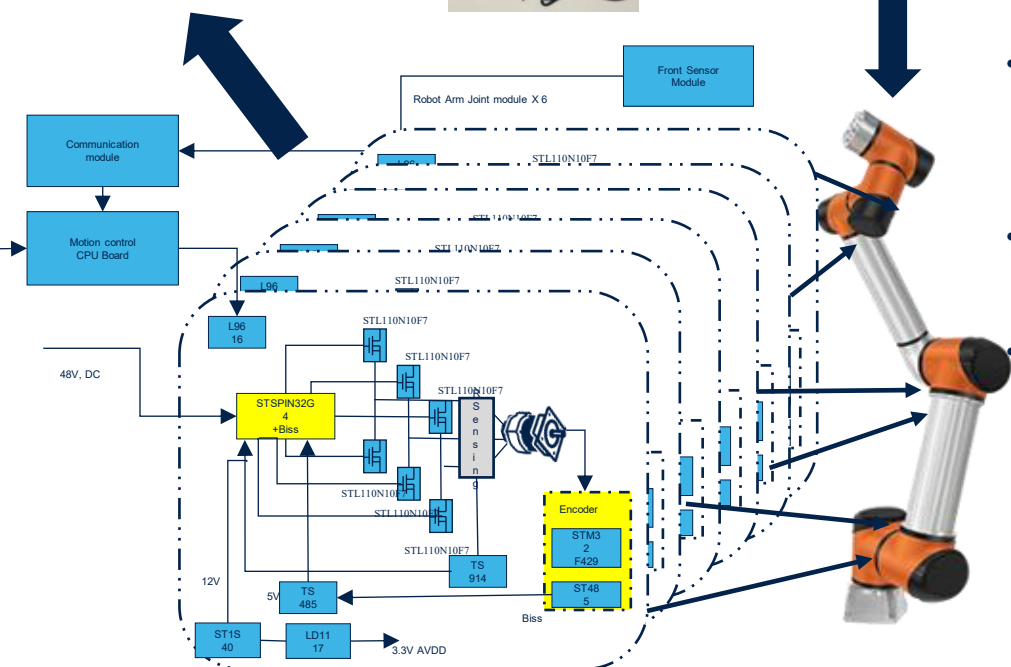
- Input DC voltage: 24~48VDC;
- Support CAN, RS485(Modbus) interface;
- Support Biss interface for Absolute encoder, this interface is very important for robotics application.
- Use STL20N6F7 as MOSFET to drive the motor;
- Use ST total solution including GPA power supply and amps

ST BOM

P/N	Quantity	W AUP \$
STSPIN32G4	1	3.6
STL20N6F7	6	1.01
TSV792IST	2	0.98
L9616	1	1.1
ST4485EBDR	3	1.12
Encoder: STM32F429	1	5
L7987LTR	1	2.5

Progress

- ↓ Feasibility Study
- ↓ Device Selection
- ↓ Schematic Design
- ↓ PCB Layout Design
- ↓ Prototype
- Test & Debug
- Design Fix
- Go to Market
- Pilot Run
- Mass Production3



Total ST Bom cost: 23.58\$
For each Joint Module;

23.58\$ X 6Unit=141.48\$ for
each robot Arm;



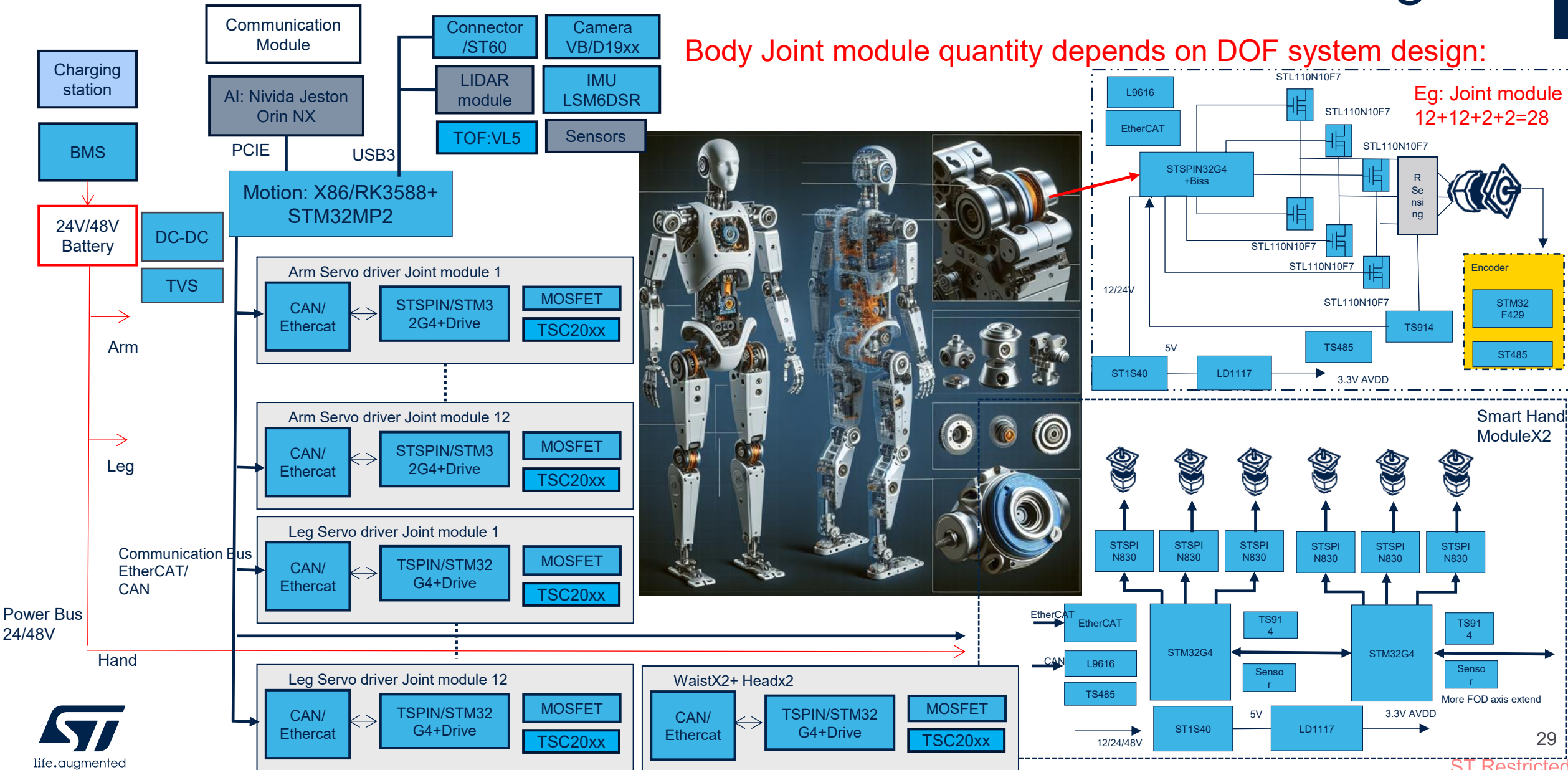
ACC+ MCCC team works



ST Humanoid Robot Block diagram

Body Joint module quantity depends on DOF system design:

Eg: Joint module
 $12+12+2+2=28$



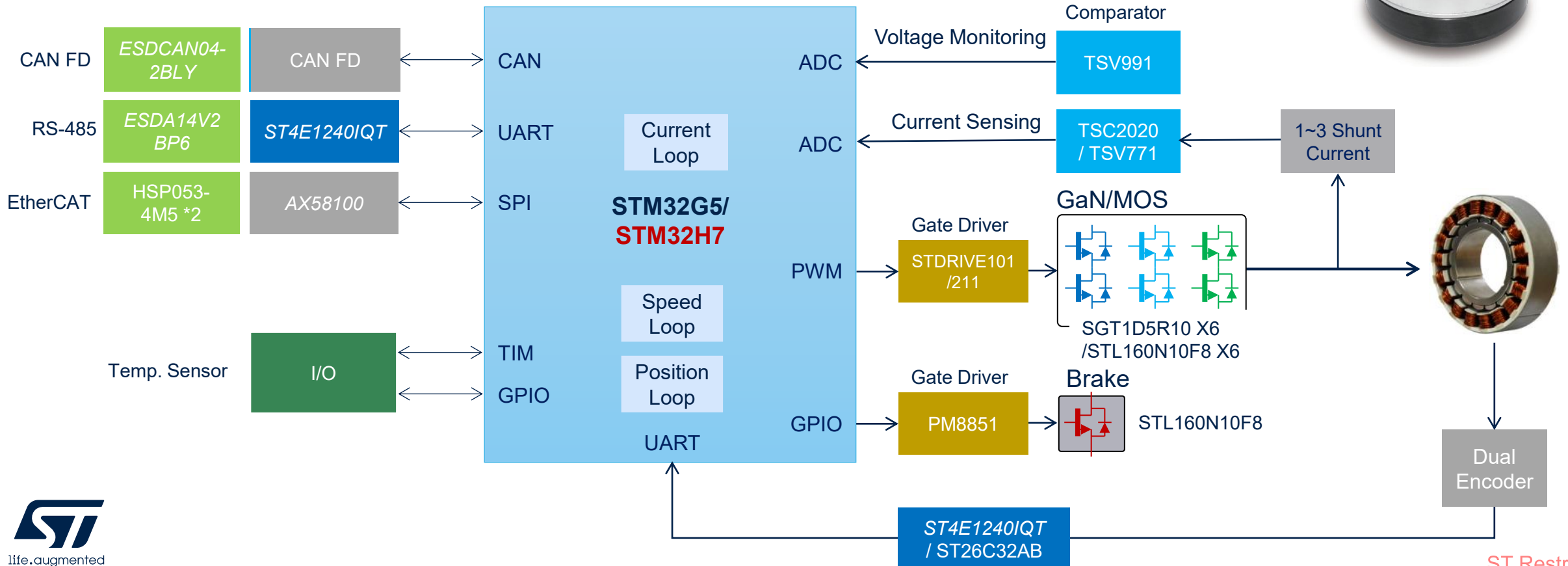
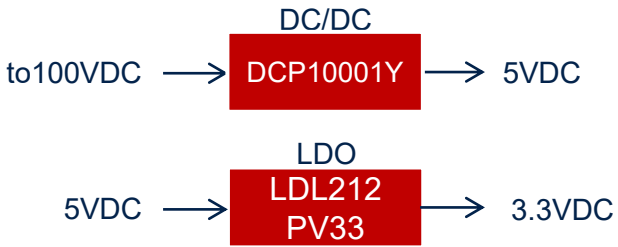
1: Discrete MCU+STDRIVE+LV GaN/MOS

100VDC MAX

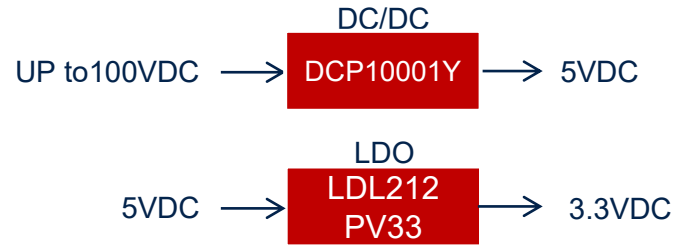


Dual Encoder

MOS/GaN	Package/ Rthja	Rdson	Motor Power/ Joint type
SGT1D5R10	FCLGA 5x6 38.1 °C/W	1.5 mΩ	2Kw above Hip/Leg
STL160N10F8	PowerFLAT 5x6 16 °C/W	3.2 mΩ	1 Kw Hip/Leg

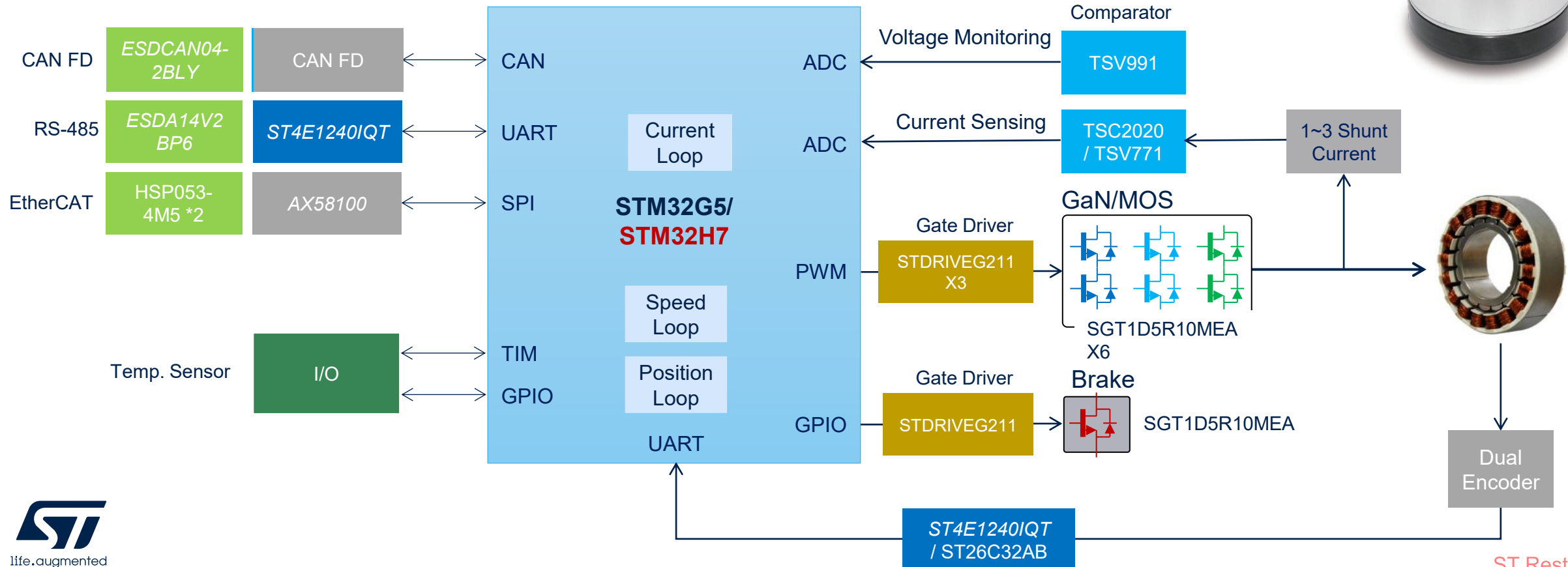


1: Discrete MCU+STDRIVE+LV GaN/MOS

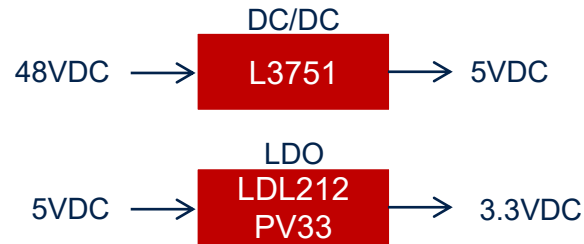


MOS/GaN	Package/ Rthja	Rdson	Motor Power/ Joint type
SGT1D5R10	FCLGA 5x6 38.1 °C/W	1.5 mΩ	2Kw above Hip/Leg
STL160N10F8	PowerFLAT 5x6 16 °C/W	3.2 mΩ	1 Kw Hip/Leg

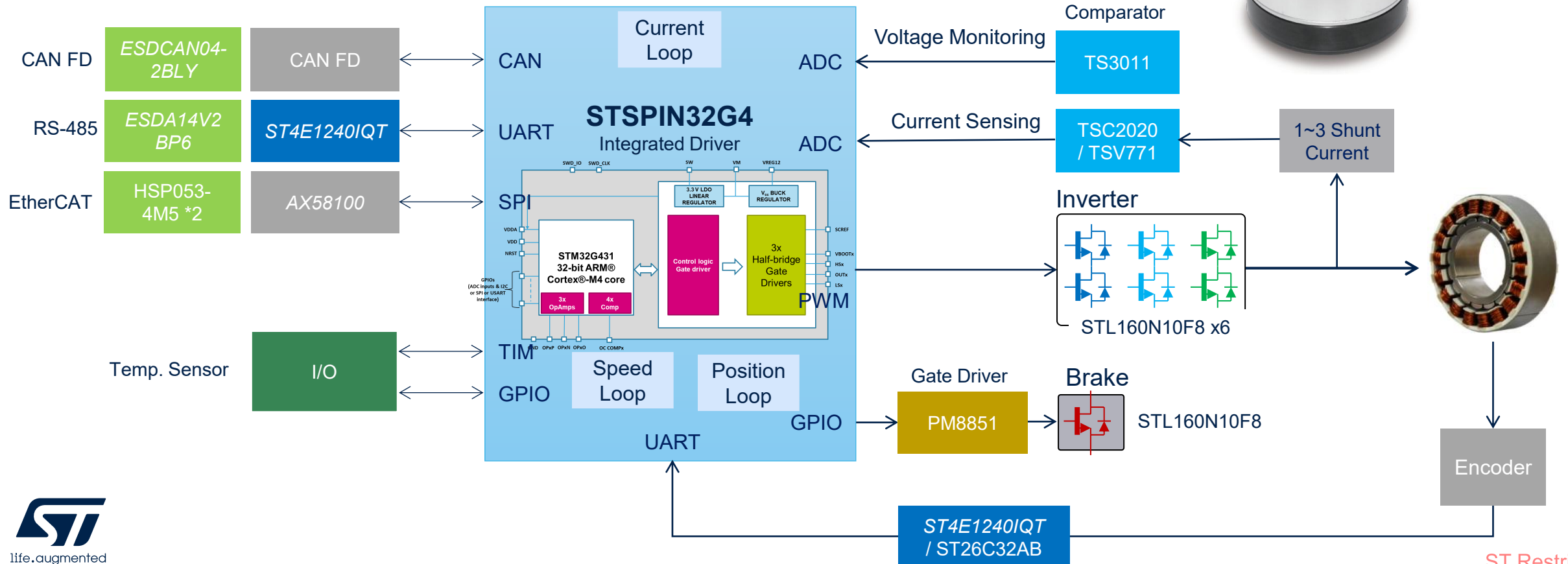
100VDC MAX



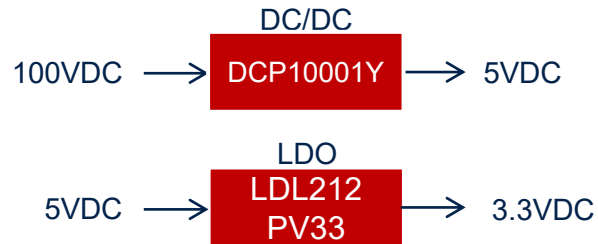
2: Integrated MCU&Driver+MOSFET



75VDC MAX

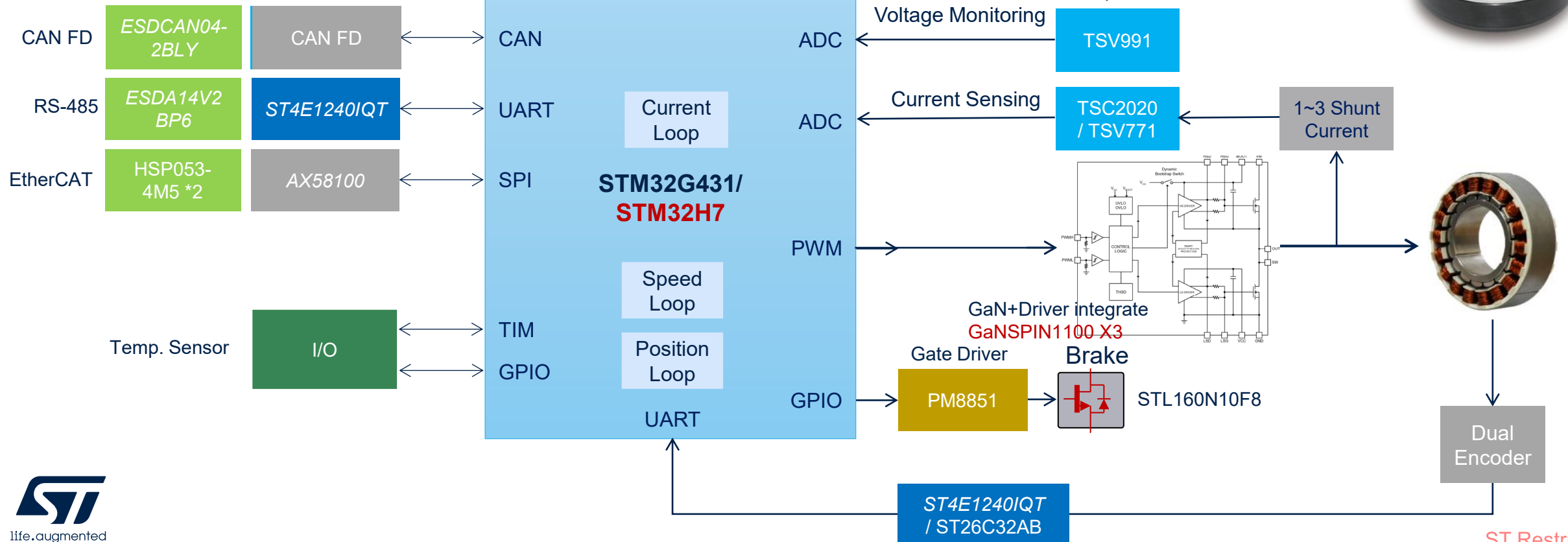


3: MCU+ Integrated: STDRIVE+LV GaN



MOS/GaN	Package/ Rthja	Rdson	Motor Power/ Joint type
SGT1D5R10	FCLGA 5x6 38.1 °C/W	1.5 mΩ	2Kw above Hip/Leg
STL160N10F8	PowerFLAT 5x6 16 °C/W	3.2 mΩ	1 Kw Hip/Leg
GaNSPIN1102	LGA 5x6.5 59 °C/W	2.4 mΩ	500W Arm/Head

100VDC MAX



Automation Competence Center Success Story

All in one robot Joint

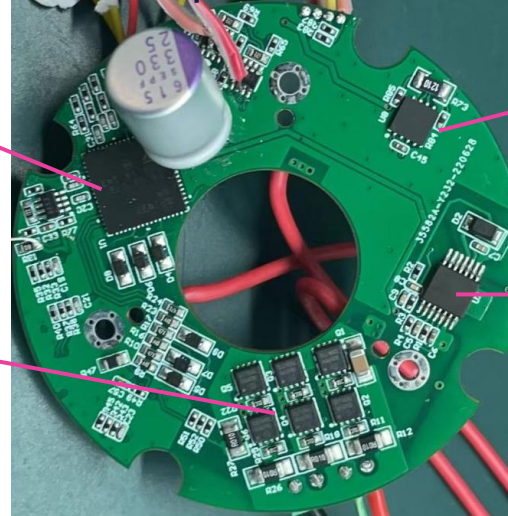
All in one Joint module

STSPIN32G4 MCU+Driver

STL110N10F7 *6
MOSFET

LD1117, LDO

Top View

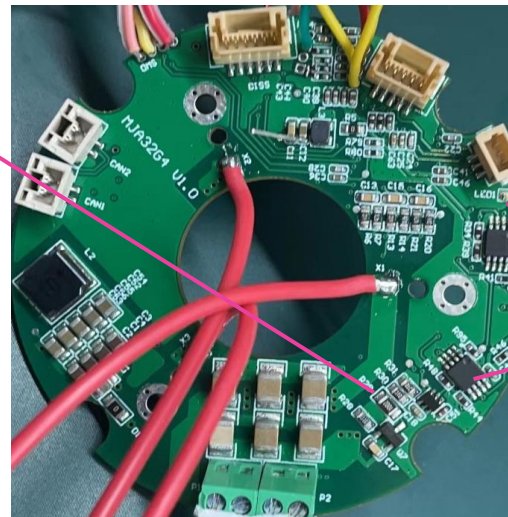
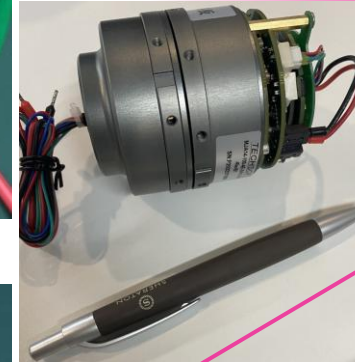


DCDC: ST1S40 1pcs

OPAMPS TS914AIDT 1pcs

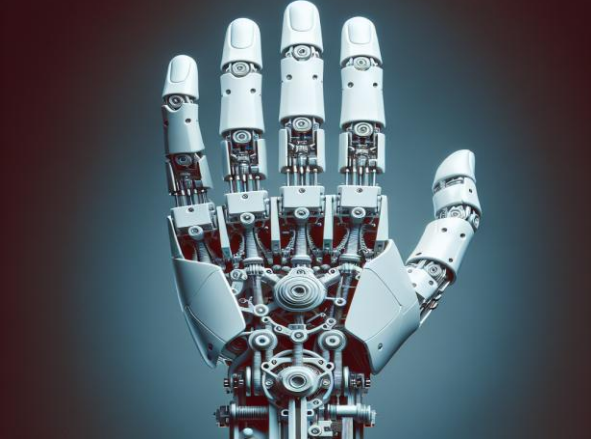
RS485 ST485EBDR 1pcs

FD CAN transceiver

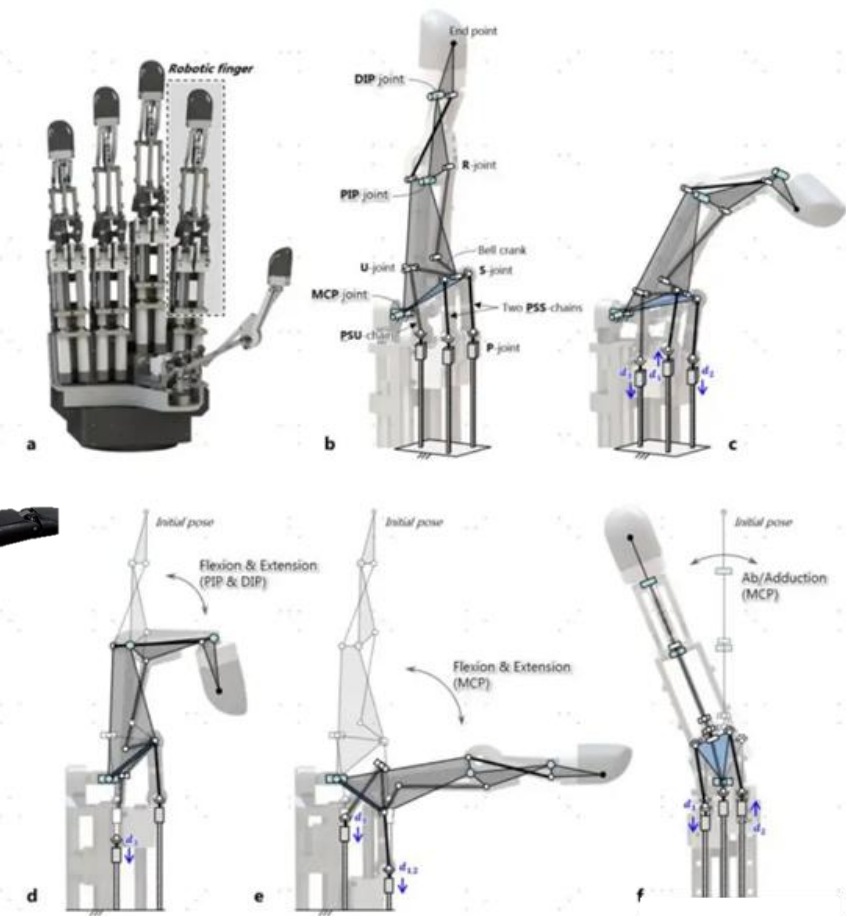
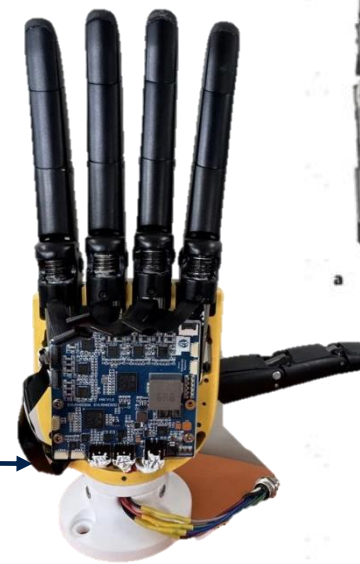
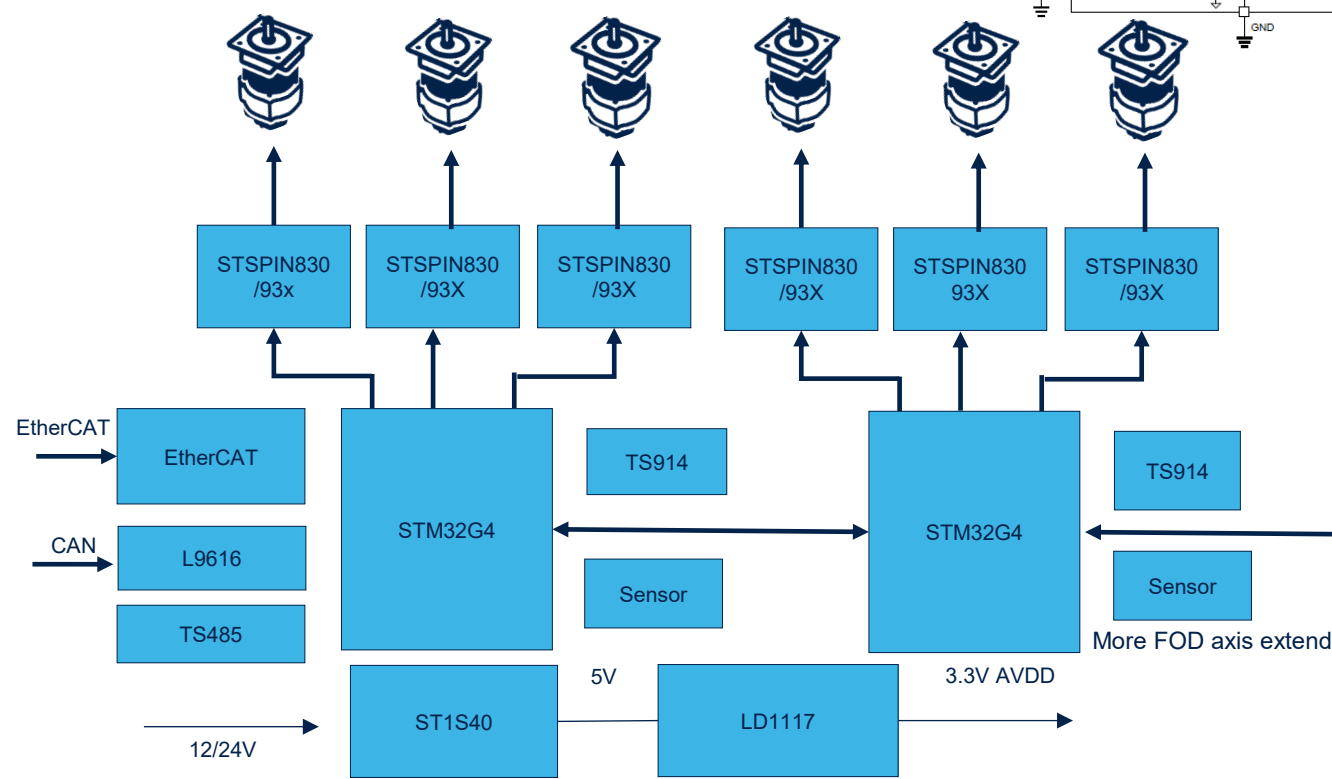
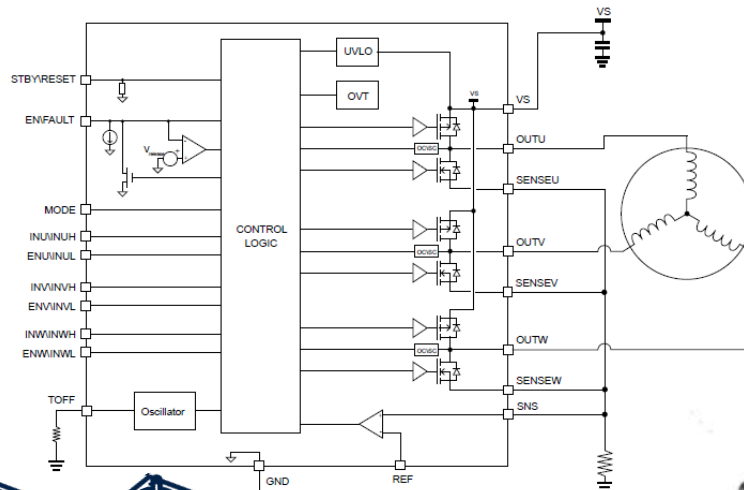


Bottom View

Compact size collaborative robot Joint Module solution



Smart Hand





6-Joint Robotic Hand Solution PCBA

New Solution

STM32G473VEH6

DCP3601NDMR

STM32G473VET6

ST4E1240IQT

ST4E1240IQT

STSPIN830/93x
* 6

LDL212PV33R

STL110N10F7

STL110N10F7

L3751PUR

SMBJ30CA

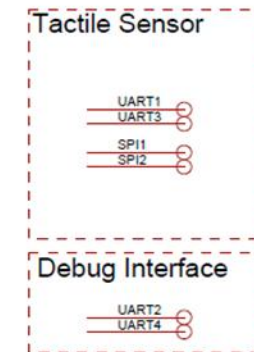
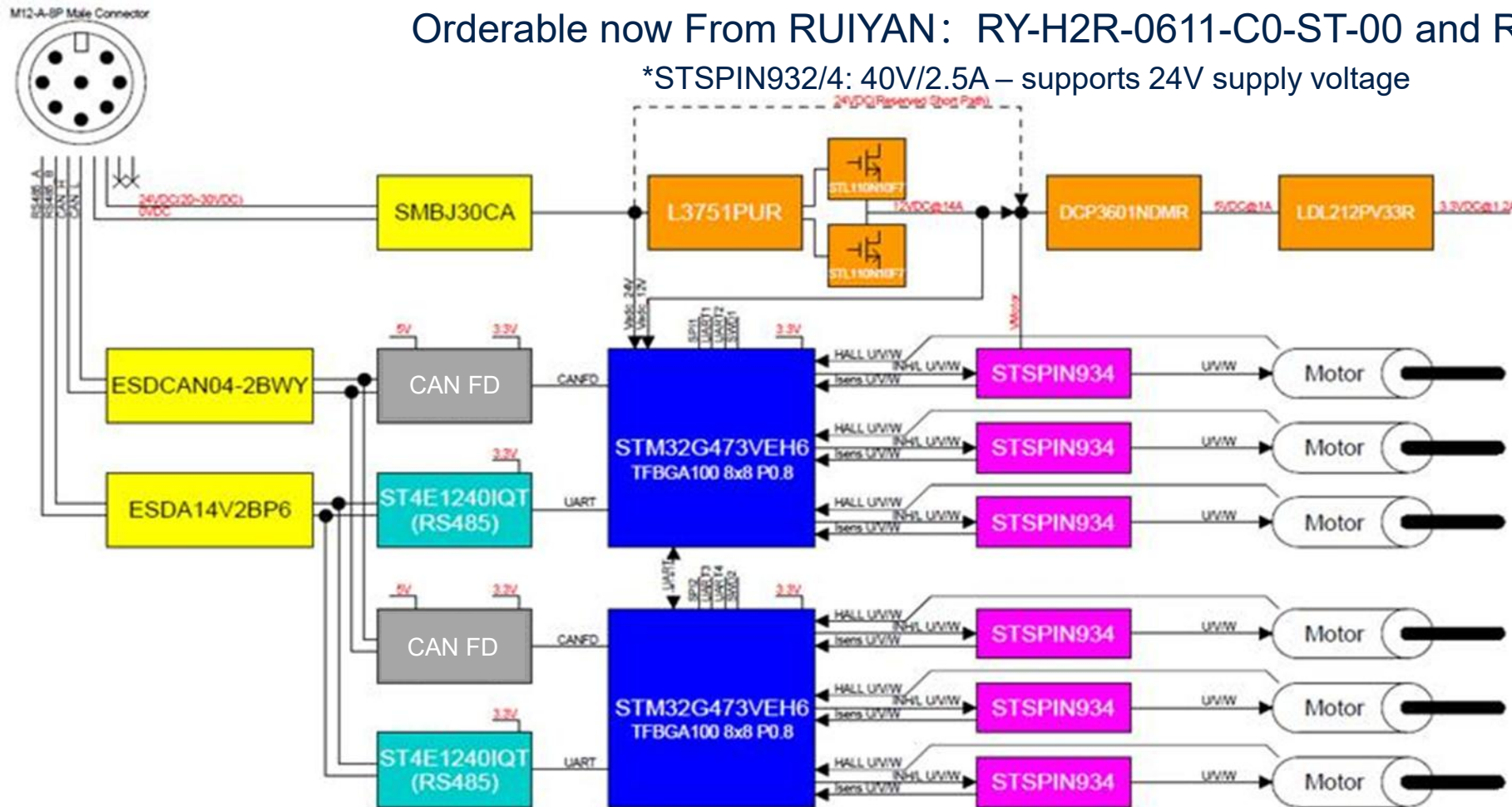


Robotic Hand : Full System Reference Design

6-Servo Robotic Hand: Reference Design (mecha platform in dev't with partner)

Orderable now From RUIYAN: RY-H2R-0611-C0-ST-00 and RY-H2L-0611-C0-ST-00

*STSPIN932/4: 40V/2.5A – supports 24V supply voltage

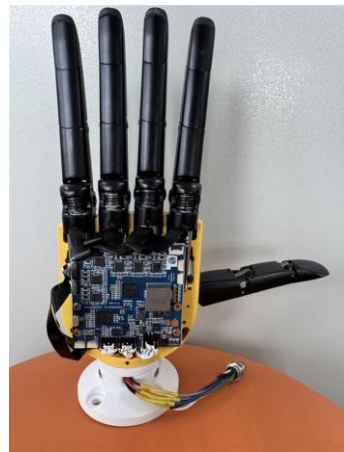


- Protection
- Power
- Transceiver
- Controller
- Driver

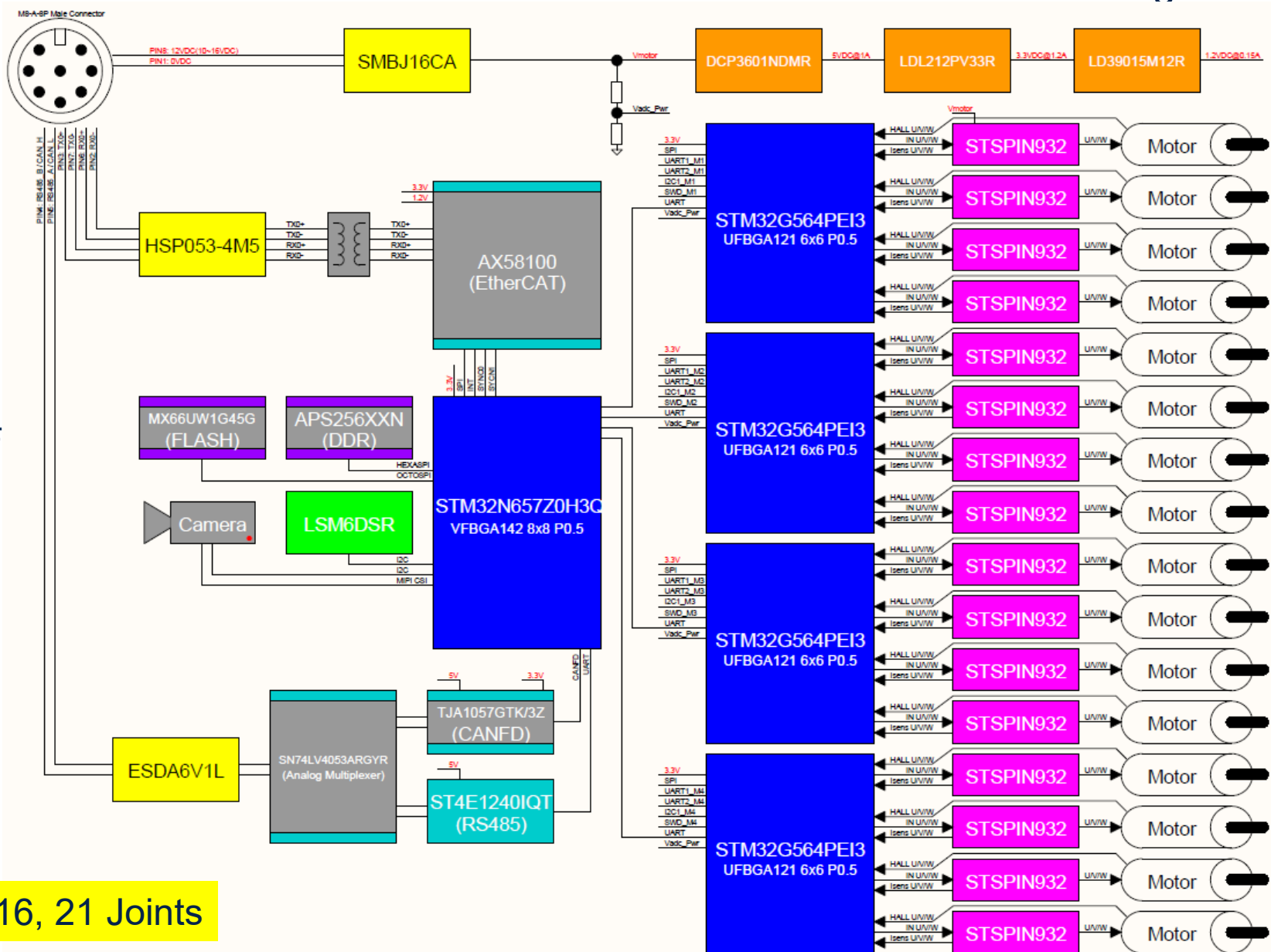
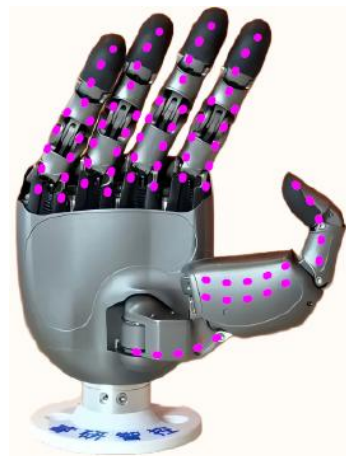


Robotic Hand

Starting Point: Hand module



From 6-DOF to 16-DOF



24VDC

15 / 16 Joint

IMU

STM32N6

Camera Module

EtherCAT

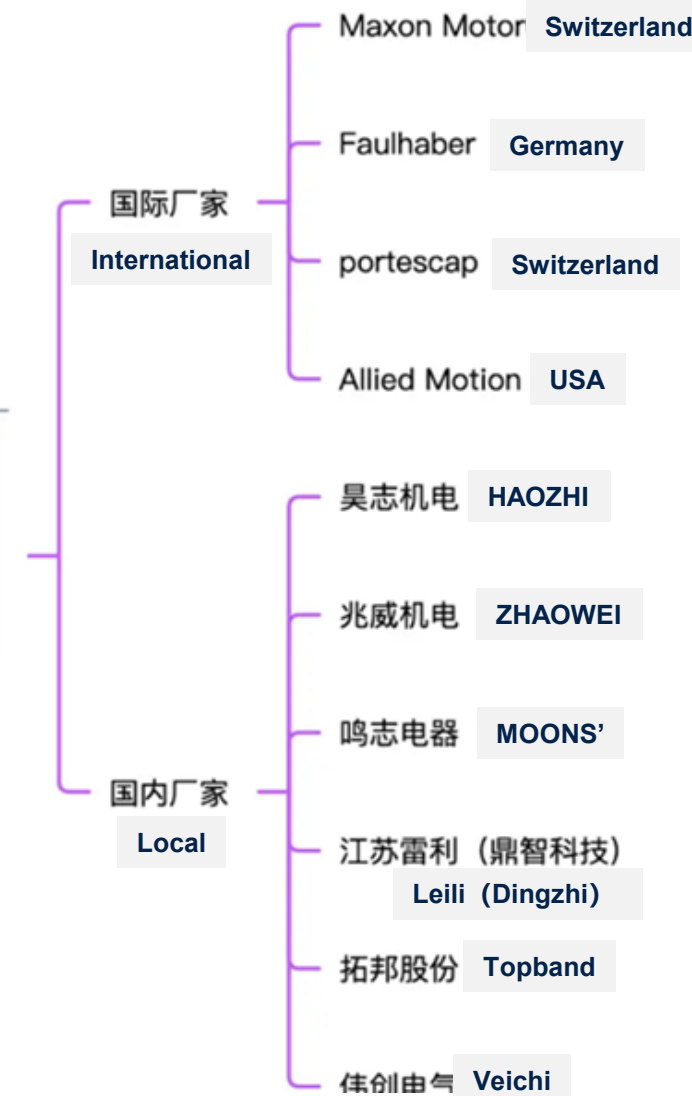
CAN FD

RS-485

6, 15, 16, 21 Joints

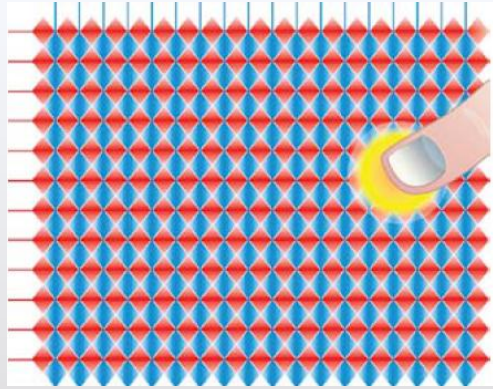
Mechanical Vender

	Figure	DOF	Weight (kg)	Load (kg)	Size	Connect	Joint
Shadow		21	4.3	5	265*84	ROPE	21
SVH		20	1.3	0.85	242.5*92	ROD	20
IH2		15	0.64	5	213*102	ROD	15
因时		6	0.54	1		ROD	12

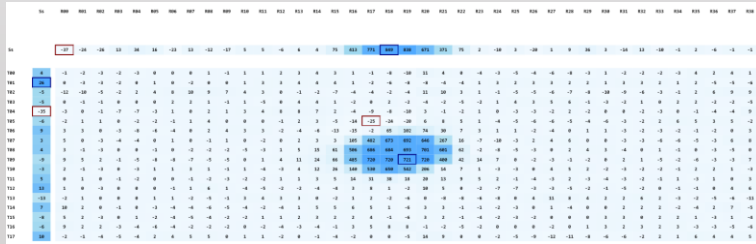


FingerTip Capacitive Sensing

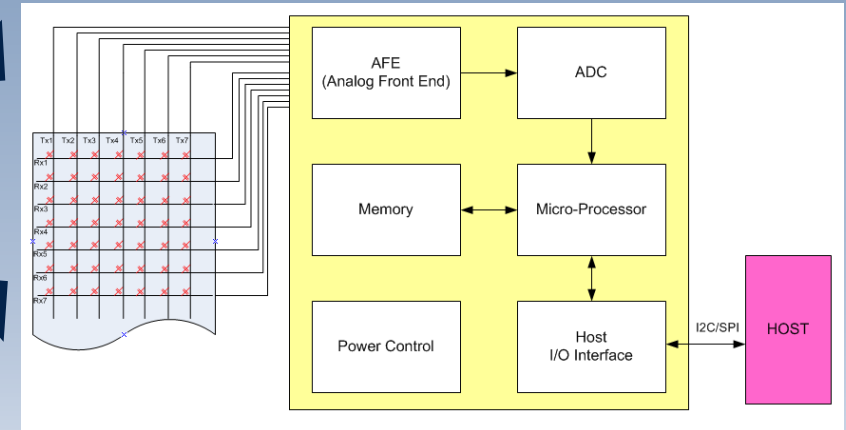
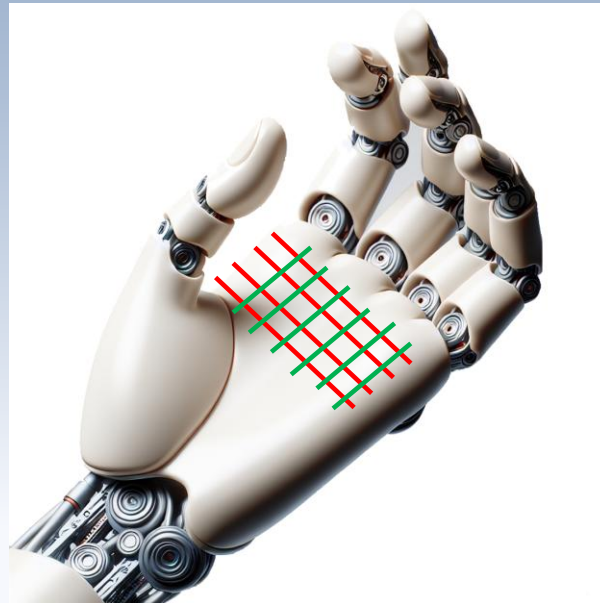
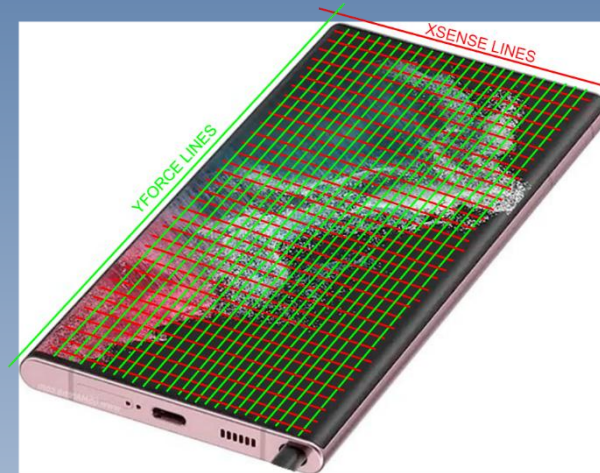
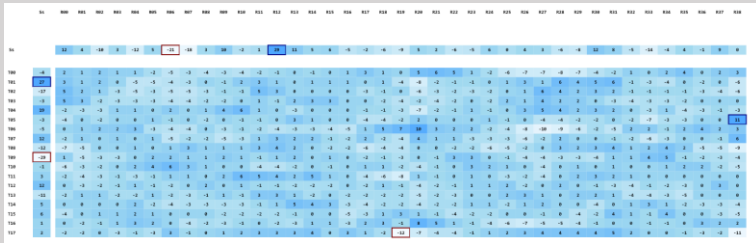
ST FingerTip support Capacitive tactile Sensing



Touch



No-Touch

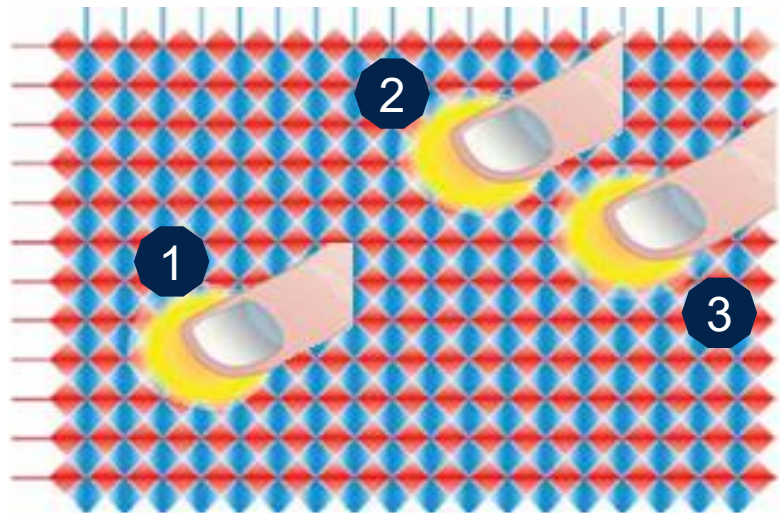


ST FingerTip
Touch IC

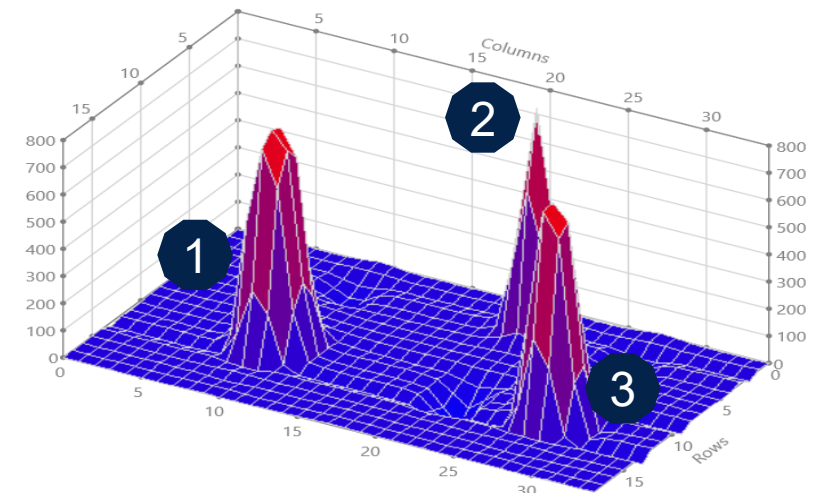
Versatile Sensing method

Mutual Scan / Self Scan

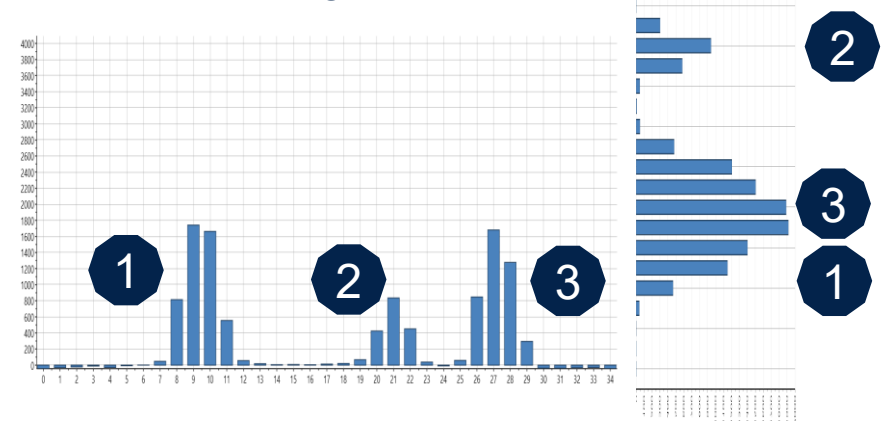
Touch Sensor & Finger



Mutual Sensing



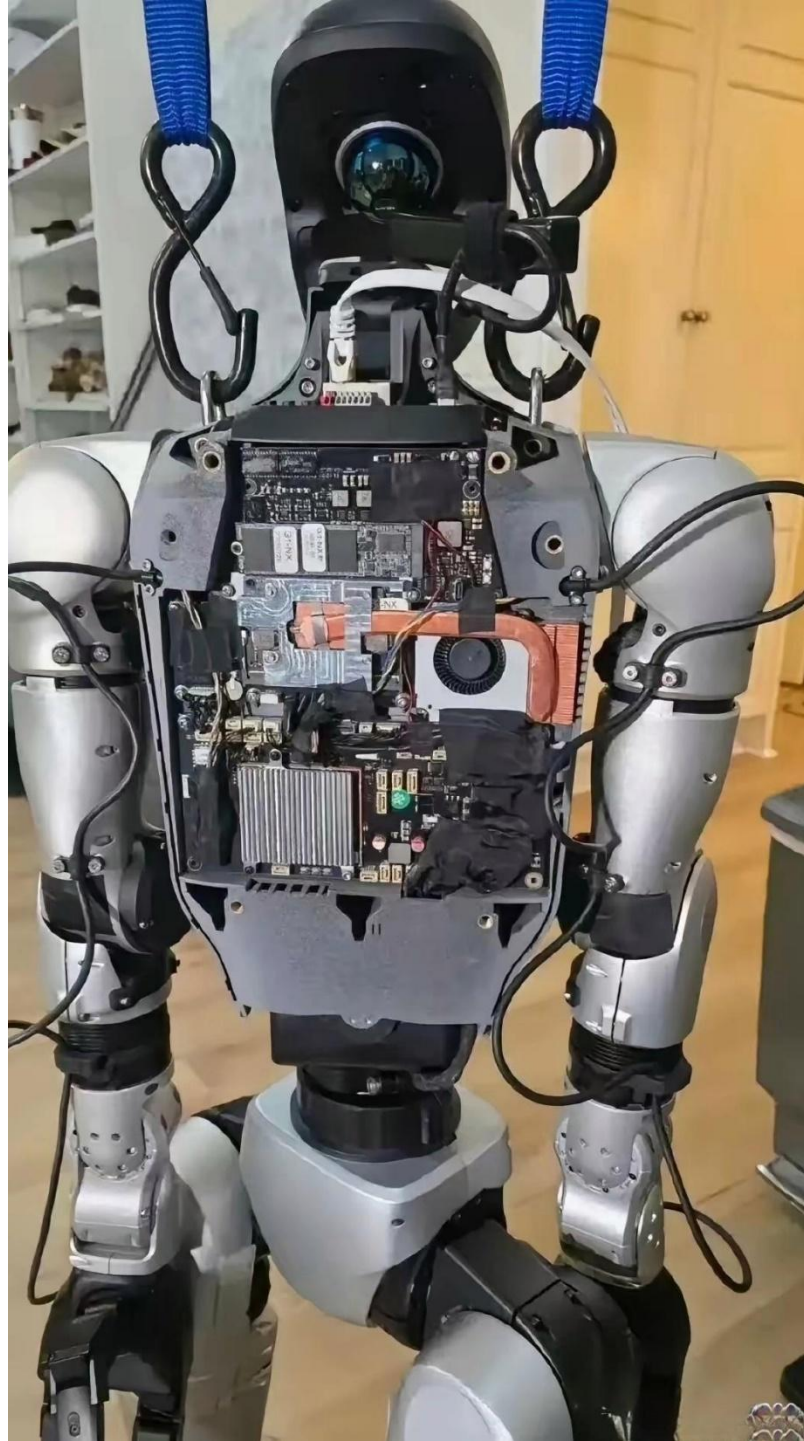
Self Sensing

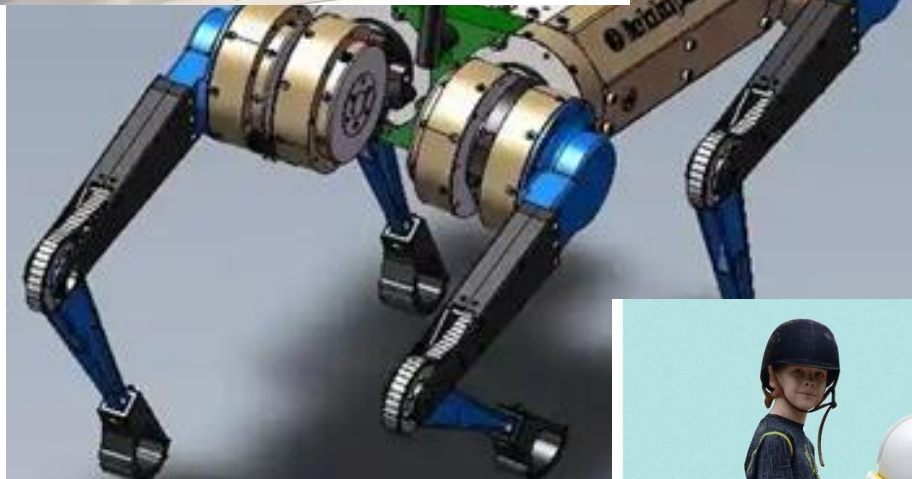


FingerTip Products Comparison

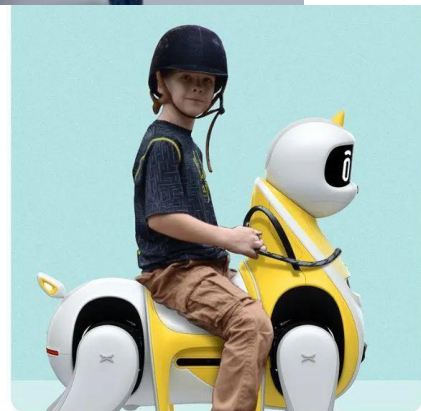
	Alix FTM5CU56A	SalixPlus FST2BA61C	Spruce FST2CA78Y	Angsana FST3CA60Y	Angsana FST3CA67Y	Casuarina FT3CA84Y	Osier and Osier-L FST4CA66Y /FST4CA84F	Remark
Tech node	TSMC 90nm	TSMC 90nm	TSMC 90nm	E40ULP 40nm	E40ULP 40nm	E40ULP 40nm	E40ULP 40nm	
Core clock	64MHz	96MHz	96MHz	150MHz	150MHz	150MHz	192MHz	
Core	M3	M4 SIMD	M4 SIMD	M4 SIMD	M4 SIMD	M4 SIMD	M33 SIMD	
Main Flash/RAM	128KB/96KB	192KB/128KB	192KB/160KB	224KB/160KB	224KB/160KB	224KB/160KB	448KB/224KB	
Tx, Rx channels	22Tx, 34Rx	21Tx, 40Rx	36Tx, 42Rx	18Tx, 42Rx	22Tx, 45Rx	39Tx, 45Rx	40Tx, 44Rx	
Capacitive Load (Typ)	200pF	500pF	500pF	500pF	500pF	500pF	840pF	
Package	BGA84 5x5	BGA87 7x4.2	BGA117 7x5	BGA87 7x4.2	BGA100 5x5	BGA117 7x5	BGA100 5x5 & BGA121 5.5x5.5	
TX driving scheme	1x,4x	1x,4x,8x	1x,4x,8x	1x,4x,8x,16x	1x,4x,8x,16x	1x,4x,8x,16x	CDMA	
Charge pump	6V	6V	6V	6V	6V	6V	-3V to 6V	

Unitree Robot

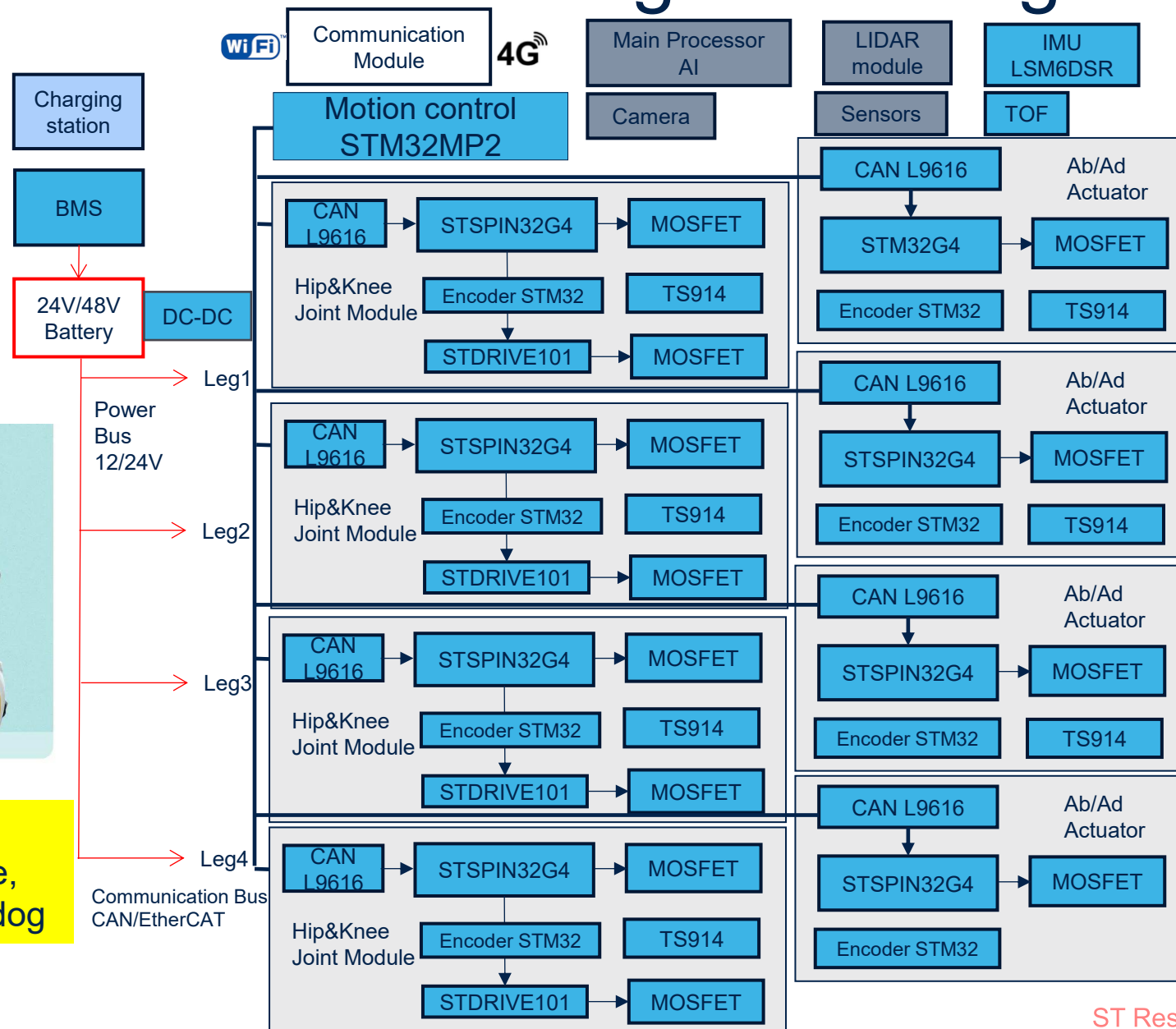




3 Joint for Each leg:
Abdomen, Hip, Knee,
12 Motors for each dog



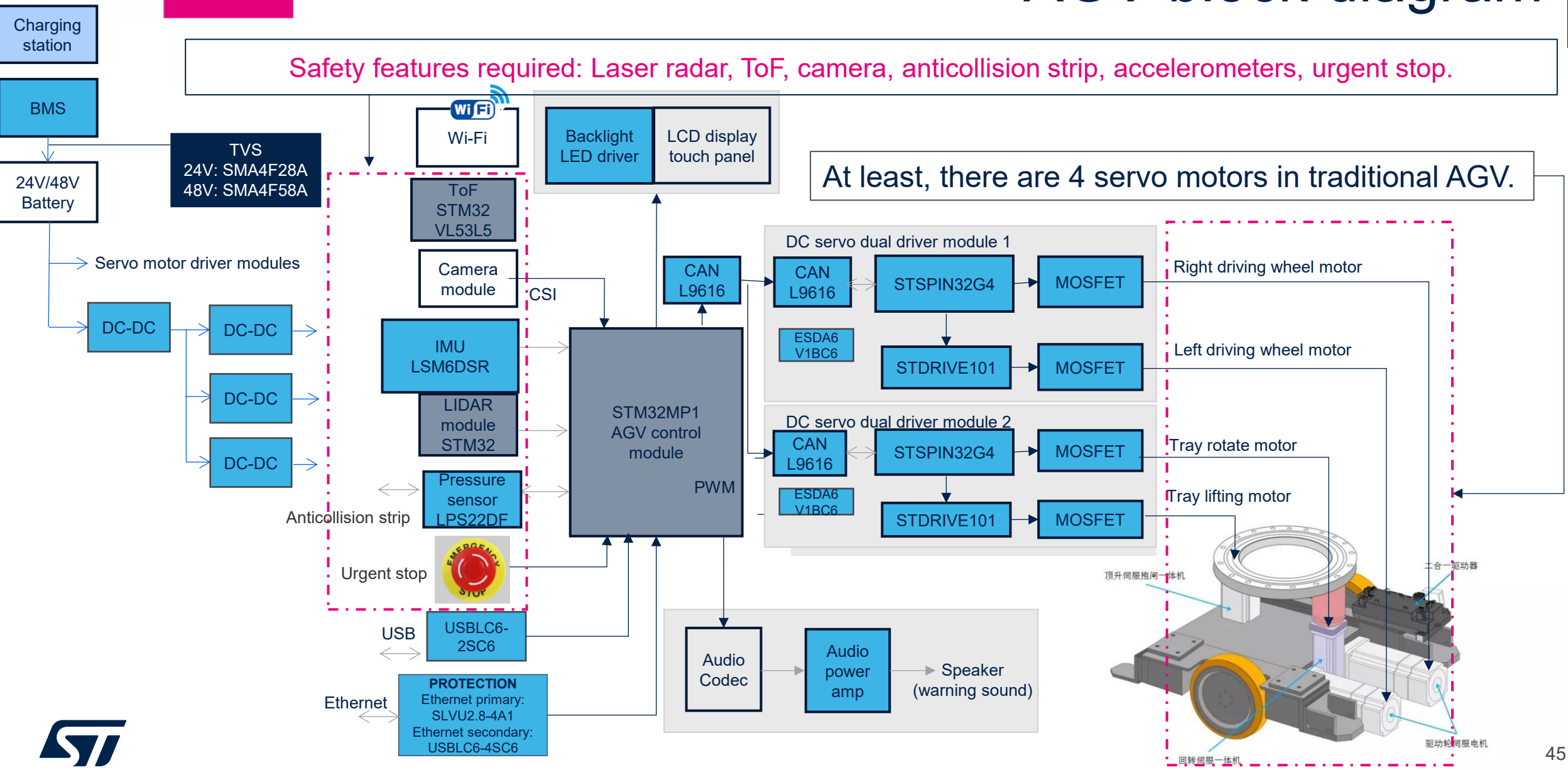
Robot Dog Block diagram



AGV block diagram

Safety features required: Laser radar, ToF, camera, anticollision strip, accelerometers, urgent stop.

At least, there are 4 servo motors in traditional AGV.





Typical AGV servo solution in market

Typical Spec for dual driver:

- Low voltage servo:
- Input Voltage: 20~70V DC
- Power level: 100W~1000W
- Dual Axis Servo Supported
- Support interface:
-Modbus\Canopen\EtherCAT
\RS485

Two in One Motor Driver

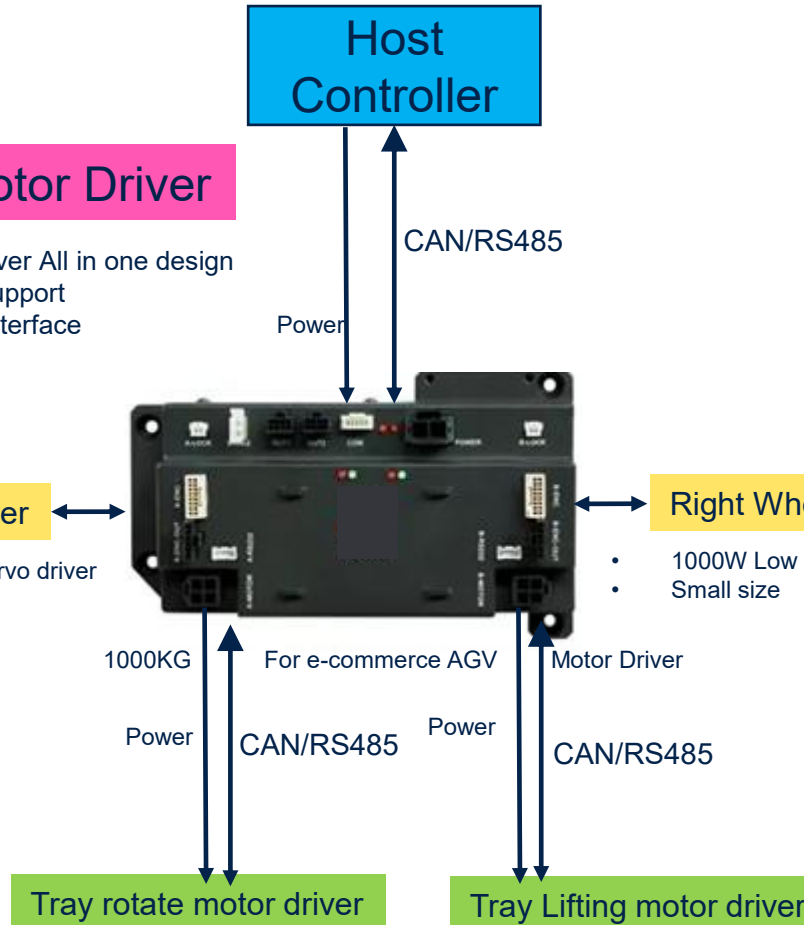
- One Controller Two driver All in one design
- STO Function safety support
- Support CAN/RS485 interface

Left Wheel Motor Driver

- 1000W Low Voltage Servo driver
- Small size

Right Wheel Motor Driver

- 1000W Low Voltage Servo driver
- Small size



- One Controller Two driver All in one design;
- 600W, 1500~3000RPM/
- STO Function safety support
- Support CAN/RS485 interface



Wireless Motor Diver

集成全60 GHz射频收发器的紧凑型解决方案



60GHz 无接触连接
-40,+105°C

- BGA 2.2x2.2 mm²
- 44 mW Tx, 27 mW Rx
- 3.5uW in Sleep

高达 6.25 Gbps

灵活的天线配置



可支持的接口类型

- Ethernet
- Display Port
- MIPI*
- VbyOne
- PCIe/SATA
- eUSB2
- UART/I2C/SPI/GPIO

* 通过接口转换芯片

自动化
测试工厂

无孔消费
电子设备

智慧工业
机器人

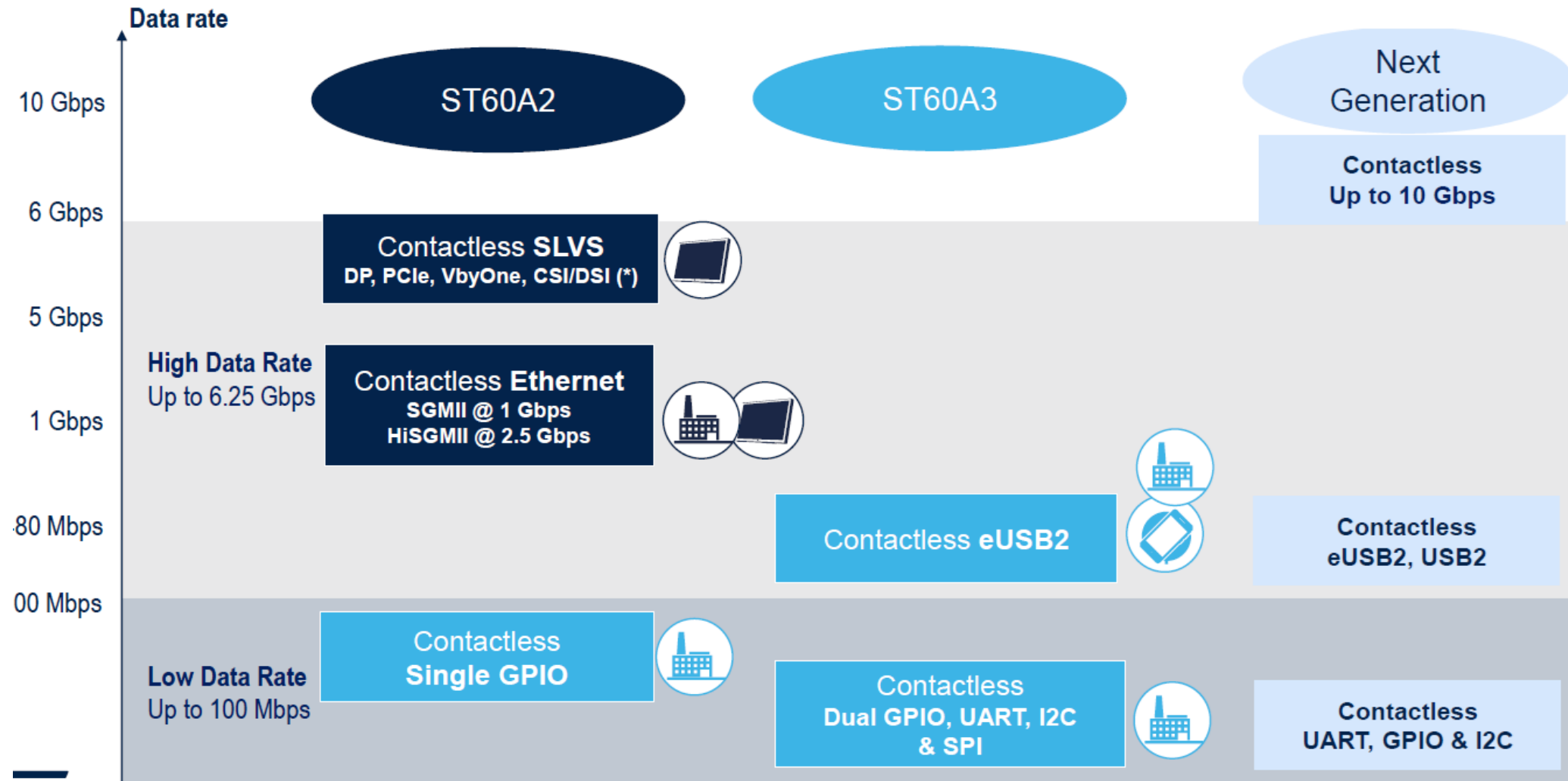
LED
视频墙

自动驾驶

无人机

非接触式
医疗设备

ST60 Products



Rosenberger solution for Robotics ARM



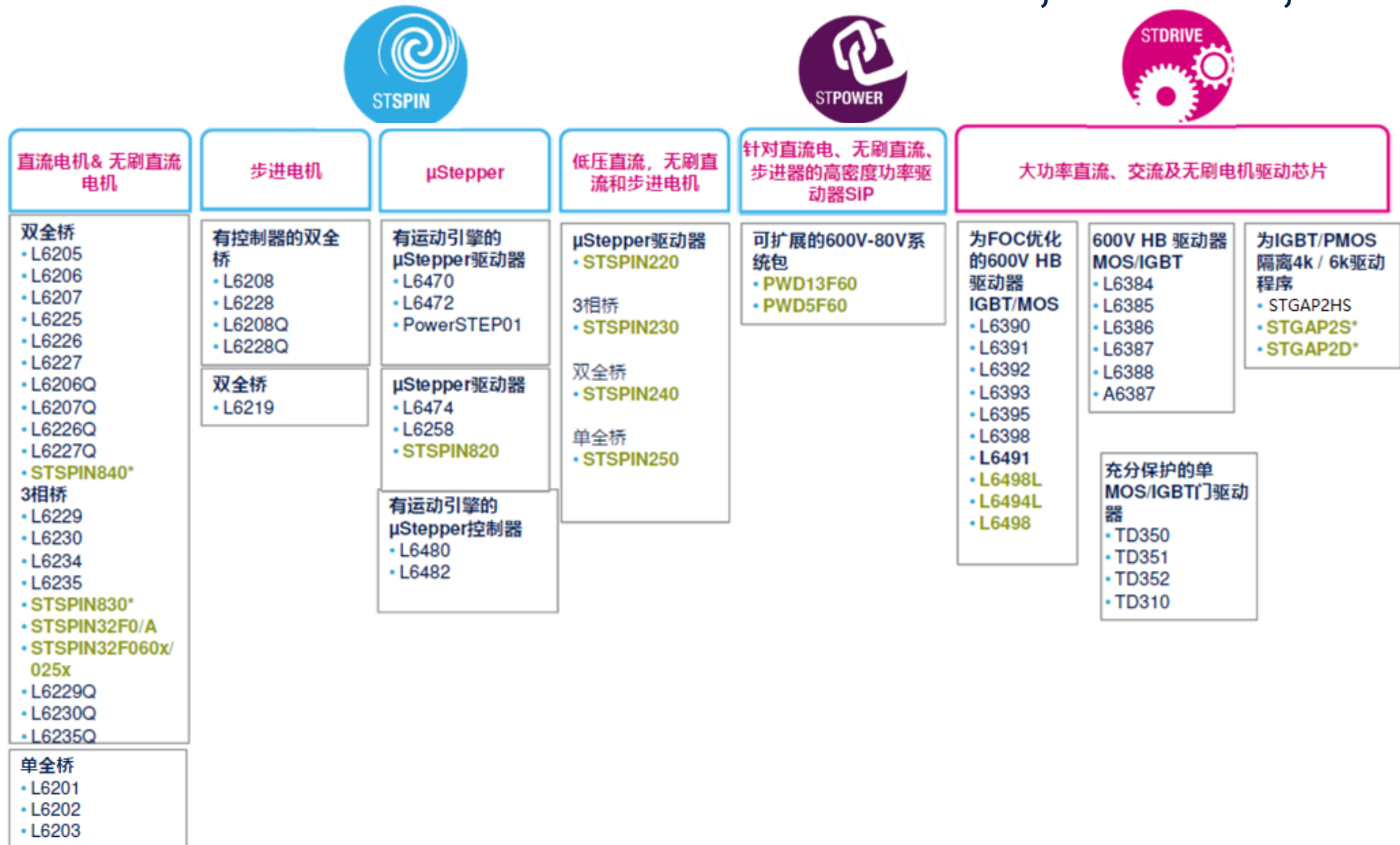
Contactless Data(ST60) & power transmission

Rosenberger Proximity Connectors (RoProxCon)



- Products that benefit from contactless technology are becoming increasingly popular
- Contactless transmission opens up a variety of new applications

ST SPIN, Drivers, Power





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